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**ENVIRONMENTAL CONTEXT OF FOREST BIOMASS
EXPLOITATION FOR BIOENERGY**

Ph.D. Thesis

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Study program: Forest Ecology

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.....
Egor Selivanov

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development for climate change mitigation, while highlighting the need for harmonized environmental and economic assessment.

Keywords: climate change; carbon neutrality; pellets; life cycle assessment; ecosystem services.

klimatu a zároveň zdůrazňují potřebu harmonizovaného environmentálního a ekonomického hodnocení.

Klíčová slova: změna klimatu; uhlíková neutralita; pelety; hodnocení životního cyklu; ekosystémové služby.

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List of publications

This thesis is written in the form of a collection of published articles. Articles I, II, and III are published in peer-reviewed journals with an impact factor. Article IV has been submitted for publication.

List of publications related to this thesis:

Publication I. Selivanov E, Cudlín P, Horáček P (2023). Carbon neutrality of forest biomass for bioenergy: a scoping review. *iForest* 16: 70-77.

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Publication II. Selivanov E., Cudlín P. (2025): Life cycle assessment of residential heat production from wood pellet combustion in the Northwest region of Russia. *Journal of Forest Science* 71(4): 182-194.

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Chapter 1: Introduction

1.1. Thesis structure

This thesis is organized into six chapters. Chapter 1 outlines the overall framework of the thesis and consists of the description of the context of the study and current state of knowledge, thesis rationale and logic flow, research goals, objectives, and hypotheses. Chapter 2 focuses on the methodological approaches used to conduct the study. This thesis is written in the form of a compilation of published peer-reviewed articles. Therefore, Chapters 3, 4, 5, and 6 include the articles published in peer-reviewed journals. Additionally, each published article is prefaced by a short summary of the main results and the contribution of the author. Chapter 7 features the overall conclusions of the study, followed by the list of references cited throughout the thesis.

1.2. Context and current state of knowledge

The global impacts of climate change (CC), caused mostly by anthropogenic greenhouse gas emissions, demand rapid action. According to Statista.com (2025), the average annual concentration of carbon dioxide (CO₂) in 2024 reached a level of 424 parts per million and is expected to increase. As pointed out in the Paris Agreement, it is crucial to reduce the amount of carbon emissions and limit the rise of global temperature “well below 2°C” (UNFCCC, 2015). Since the main source of CO₂ emissions is combustion of fossil fuels (IPCC, 2022b), CC mitigation actions should focus on the transition from fossil fuels to alternative renewable energy sources. Alternative energy sources include, but are not limited to, solar power, wind power, hydropower, and geothermal energy. Another renewable energy option is bioenergy.

Major international agencies, such as the International Energy Agency (IEA, 2021), International Renewable Energy Agency (IRENA, 2020), European Union’s Renewable Energy Directive (RED II, 2023), and Intergovernmental Panel on Climate Change (IPCC, 2022b) recognize bioenergy as an important alternative to fossil fuels. Currently, fossil fuels are the

most dominant source of energy production; the share of renewable energy in global total primary energy consumption is only 13% (Our World in Data, 2025), with bioenergy being the biggest renewable energy source (58% of all renewable energy consumption). Bioenergy can be produced from a diverse range of feedstocks, including agricultural residues (e.g., wheat straw, rice husk), organic waste (e.g., municipal waste, food waste), aquatic biomass (e.g., algae), and forest biomass.

Forest biomass is the most dominant bioenergy feedstock. In 2017, 69% of global bioenergy produced came from forest biomass (Kozuch et al. 2024). The widespread adoption of forest biomass as a feedstock for bioenergy is based on the following key factors:

- **Renewability.** Unlike fossil fuels, which require millions of years to form and are thus irreplaceable once extracted, forest biomass can regrow in decades under sustainable forest management practices.
- **Abundance and availability.** Forests cover around 30% of the planet's land surface (FAO, 2020). Unlike conventional fossil fuels and some more technologically advanced renewable energy options (e.g., hydrogen or solar power), forest biomass offers comparatively straightforward feedstock acquisition and energy conversion processes.
- **Versatility.** Forest biomass is unique as it can be converted into all three forms of bioenergy: solid biofuels (e.g., wood pellets, briquettes), gaseous biofuels (e.g., syngas, biomethane), and liquid biofuels (e.g., bioethanol). These can be produced through various thermochemical (e.g., direct combustion, pyrolysis) and biochemical (e.g., anaerobic digestion, fermentation) processes.
- **Carbon neutrality (CN).** According to the basic concept of CN, the amount of CO₂ emissions resulting from biomass combustion is equal to the amount of CO₂ sequestered during plant growth. However, the status of CN of forest biomass is currently debated in the scientific

community (Norton et al. 2019). One of the main objectives of this thesis is to analyze the concept of CN of forest biomass and identify the underlying assumptions and key limitations.

Despite the ongoing scientific debate regarding the CN status of forest biomass, it remains a widely utilized energy source. To better understand the environmental impact of forest biomass used for bioenergy as a solution for CC mitigation, numerous studies have been conducted utilizing the Life Cycle Assessment (LCA) approach. LCA is a standardized methodology used to systematically analyze the environmental impacts of a product or system throughout its entire life cycle, from raw material acquisition to end-of-life disposal (ISO, 2006a). In the context of forest-based bioenergy production, numerous studies have used LCA to analyze the environmental impact of the utilization of various kinds of biofuels. Such LCA projects focus on different aspects of bioenergy production, including impacts of different feedstocks (Sjølie and Solberg 2011; Morrison and Golden 2016), conversion methods (Carotenuto et al. 2022), and transportation options (Pierobon et al. 2015; Vera et al. 2019). Various studies investigate bioenergy generation via the utilization of different biofuel types, such as wood chips (Hammar et al. 2017; Klavina et al. 2017) and briquettes (Alanya-Rosenbaum et al. 2018; Sahoo et al. 2021). According to a systematic review on forest-derived solid biofuels conducted by Musule et al. (2021), wood pellets are the most studied and utilized type of biofuels. Numerous LCA studies have been carried out to evaluate the environmental impact of pellet production and combustion. For example, Pergola et al. (2018) compared the environmental impact of pellet production from roundwood logs to pellet production from sawmill residues (mainly sawdust). Ahmadi et al. (2020) analyzed the difference between power generation in Canada from gasification of woodchips originating from softwood harvesting residues to gasification of pellets, combustion of pellets, and pyrolysis of woodchips. Furthermore, due to the increased production of pellets for combustion and higher heating efficiency compared to conventional firewood, pellets are becoming more commonly used for household heat generation. Several LCA studies have focused on assessing the environmental impact of pellet combustion for

residential heat production. For example, Sgarbossa et al. (2020) evaluated the environmental profile of four wood pellet supply chains for heat production from different types of biomass feedstock. In Slovenia, Topić Božič et al. (2024) conducted an LCA that compared the effects of heat production via combustion of wood pellets vs. firewood at the household level.

One of the main objectives of this thesis is to utilize the LCA methodology to assess the environmental impacts of heat generation via pellet combustion in Russia, addressing the current gap in LCA studies on bioenergy production in this region. This will critically analyze the effectiveness of this bioenergy production scenario as a CC mitigation option, while testing the CN status of the biofuels utilized.

In order to comprehensively analyze the validity of bioenergy generation as a CC mitigation strategy, environmental impact assessment should be supported by economic evaluation. For a robust assessment, all analyzed ecosystem services (ES) related to bioenergy production should be evaluated. According to Millennium Ecosystem Assessment (MEA, 2005), The Economics of Ecosystems and Biodiversity (TEEB, 2010), and Common International Classification of Ecosystem Services (CICES) frameworks (Haines-Young and Potschin-Young, 2018), forest biomass exploitation for bioenergy primarily falls under the provisioning ES category, supplying renewable feedstock for energy production. Moreover, bioenergy production provides a number of additional ES, such as carbon sequestration, climate regulation, biodiversity, and others. Thorough assessment of the economic value of bioenergy production, its potential, and its trade-offs requires application of a proper valuation methodology. Therefore, another key objective of this thesis is to provide a detailed description of the available monetary valuation methods for ES assessment by conducting a scoping review. This will establish a structured framework aimed to facilitate future economic analysis of bioenergy production scenarios.

Forest biomass used for bioenergy is considered a viable alternative to fossil fuels and an attractive solution to combat CC, but its carbon neutrality status and broader sustainability are disputed. This thesis addresses these issues by (1) critically examining the carbon neutrality concept of forest biomass in the context of bioenergy, (2) conducting an LCA to evaluate the environmental impact of bioenergy generation in the Northwest region of Russia, and (3) thoroughly analyzing available methods for monetary valuation of ES. Together, these contributions can be used to inform policymakers and stakeholders regarding CC mitigation strategies.

1.3. Thesis rationale and logical flow

The articles comprising this thesis are logically interconnected, with each one building upon the previous to form a cohesive narrative to extend the understanding of sustainable bioenergy exploitation. Publication I (Chapter 3), a scoping review of the carbon neutrality concept of forest biomass, establishes the conceptual foundation by providing insight into key debates that surround the carbon neutrality status of forest biomass. Building on this, Publication II (Chapter 4), LCA of residential heat production from pellet combustion in Russia, applies the theoretical framework discussed in the first paper to a case study by quantifying the environmental impact from the combustion of wood pellets. Publication III (Chapter 5), a scoping review of ES monetary evaluation methods, shifts the focus from environmental assessment of bioenergy to economic evaluation. By identifying methods for evaluation of ES in monetary terms, this publication provides a set of tools for economic assessment, facilitating a foundation for CC-related policy decisions. Finally, Publication IV (Chapter 6), a short communication titled “Is forest biomass used for bioenergy production a viable climate change mitigation option?”, addresses the idea of carbon neutrality of forest biomass by combining key findings from the first two articles. It further discusses whether forest biomass is a viable CC mitigation option.

Together, these articles form a progressive arc: from theoretical foundations of the concept of CN of biomass (Publication I) to experimental validation through LCA of bioenergy production (Publication II) to policy-relevant monetary valuation methods for ES (Publication III) to the synthesis of the findings regarding the role of biomass in CC mitigation (Publication IV).

While this thesis provides the tools for economic assessment of ES, it does not conduct a full monetary valuation of bioenergy generation itself. This limitation arises from two key factors:

1. Data limitations: A thorough monetary valuation would require primary data that is currently unavailable.
2. Scope of the thesis: Despite recognizing the importance of economic valuation of ES, the main focus of this thesis prioritizes resolving conceptual and empirical uncertainties surrounding CN and bioenergy production.

Future research should build upon this thesis's findings by conducting a comprehensive monetary valuation of bioenergy production. The results of the environmental analysis presented in this thesis, when combined with the economic insights obtained through potential monetary evaluation of bioenergy production, will provide a robust foundation for CC mitigation policies.

1.4. Goals and objectives

The overarching goal of this thesis was to advance the understanding of the role of forest biomass exploitation for bioenergy production as an option for CC mitigation.

Objectives of the thesis:

1. Analyze the concept of CN of forest biomass in the context of bioenergy production with a focus on identifying its underlying assumptions and key limitations.

2. Apply an LCA methodology to a case study to measure the environmental impacts of heat production from the domestic combustion of woody pellets derived from sawmill residues.
3. Identify the existing methods for monetary evaluation of ES.

Completing the first objective will allow for a deeper understanding of the CN concept, providing the framework for more informed analysis of forest biomass sustainability in bioenergy production. Completing the second objective will determine how utilization of wood biomass affects the environment throughout its life cycle. Completing the third objective will provide clear solutions on how to assess bioenergy production in monetary terms, facilitating the presentation of the results to policy makers and stakeholders for more informed CC-related decision making.

By addressing these objectives, this thesis aims to provide a holistic perspective on the sustainability of forest-based bioenergy systems, integrating environmental, economic, and ecological dimensions to inform policy and practice.

1.5. Hypotheses

The overarching hypothesis of this study is that forest biomass can be used for bioenergy production as a measure to combat CC, provided its CN, environmental impacts, and economic value are rigorously evaluated.

Working hypotheses:

Hypothesis 1: There is no commonly accepted definition of CN of forest biomass, and CN status is contingent upon specific assumptions and methodological choices.

Hypothesis 2: Forest biomass used for bioenergy production within specific system boundaries has a low environmental impact and can be considered effective for CC mitigation.

Hypothesis 3: The environmental impact of forest-based bioenergy production depends on the structure and efficiency of its supply chain.

Hypothesis 4: Monetary valuation methods for ES are numerous but lack consistent terminology and require a standardized framework to improve their applicability for bioenergy-related policy making.

Chapter 2: Methodology

Due to the fact that this thesis explores different aspects of forest biomass exploitation and is presented in the form of compilation of published articles, different methodological approaches were utilized to conduct these studies.

The publications presented in Chapter 3 and Chapter 5 follow a scoping review methodology, whereas Chapter 4 utilizes a standard LCA approach. The publication presented in Chapter 6 adopts the short communication format and employs its own specific concise structure and uses qualitative synthesis approach. Below is a detailed description of methods used in each particular study.

2.1. Carbon neutrality of forest biomass for bioenergy: a scoping review

This section of the thesis describes the methodology used in the publication presented in Chapter 3 of the thesis.

A scoping review is a methodological approach designed to systematically analyze existing scientific literature with diverse study designs and methods. This approach describes the findings from different studies, making it possible to identify and analyze knowledge gaps, clarify concepts and definitions, and identify key topics within a desired field of research. In this scoping review, we utilized a framework introduced by Arksey and O'Malley (2005), recommendations from Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews (Tricco et al. 2018), and guidance for scoping review from Joanna Briggs Institute (Peters et al. 2015). A scoping review method consists of five major steps.

The first step is *identifying research questions*. The main objectives of this study were to identify the most widely used definitions of CN and to analyze and describe the key assumptions and underlying CN concepts. The research questions were:

- What is the definition of carbon neutrality?
- What are the main assumptions that carbon neutrality concepts are based on?

The second step is *identifying potentially relevant studies*. Figure 1 summarizes the study selection process. During this step, we conducted web-searches in Web of Science (WoS) and Scopus databases. By using particular key phrases (“carbon neutrality”, “carbon neutral biomass”, “carbon neutral bioenergy”, and “forest carbon neutrality”) and inclusion criteria (peer-review articles and gray literature; publications in English language; main focus on carbon neutrality of forest biomass), we were able to identify 5699 papers.

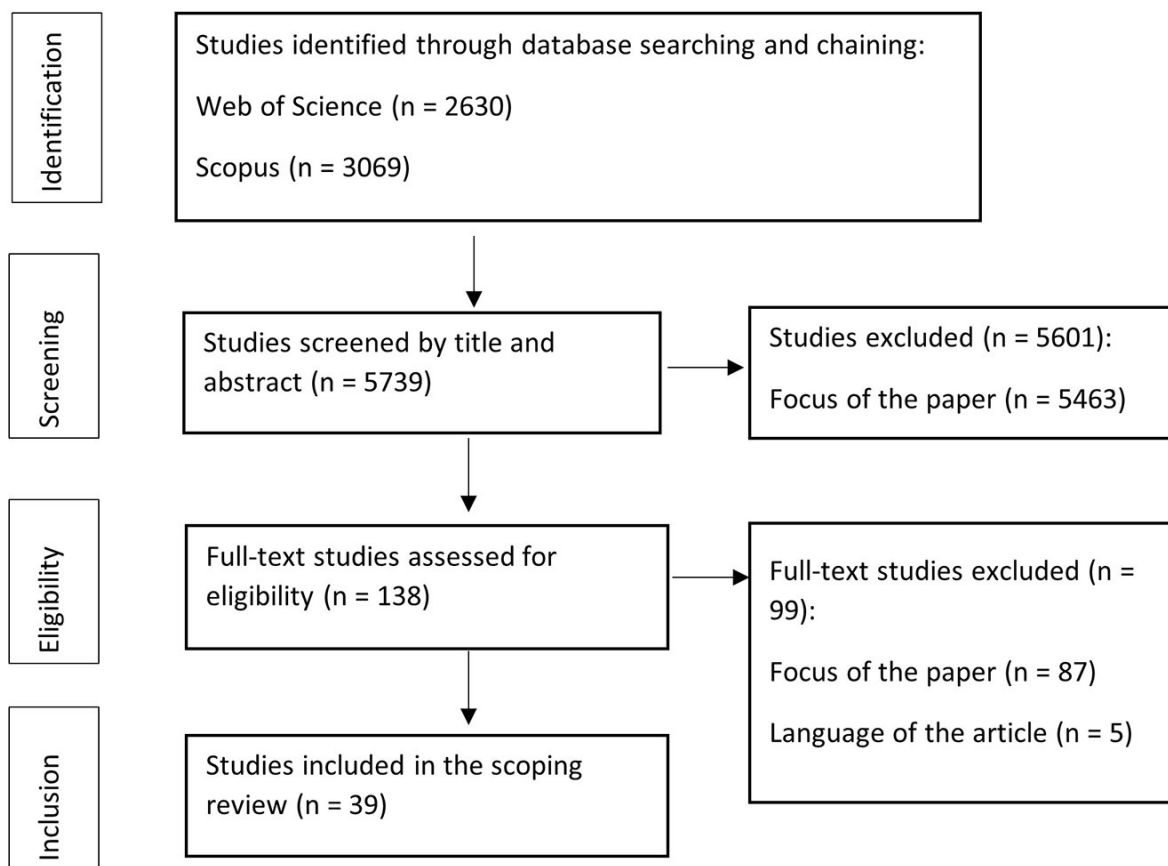


Figure 1. Flow chart of the search strategy of a scoping review

The third step is *selecting eligible studies*. During the screening of the 5699 identified articles, 5463 of them were excluded from the scoping review due to the lack of relevance to the topic based on the content, title, or abstract of screened articles. Additionally, 138 articles were

excluded as duplicates. Finally, a full-text analysis was performed on the remaining 138. Finally, 99 studies were excluded for not meeting the inclusion criteria, leaving 39 articles to be analyzed in our scoping review.

The fourth step is *charting the data*. The descriptive qualitative approach was employed to analyze the eligible articles. For this purpose, a data-charting table was created. This table included the following information: name of the author, year of publication, type of publication, goal of the study, definition (concept) of CN used in the article, comparison of fossil fuels to biomass with the focus on CN, characteristics related to the CN concept considered in the article, and general perception of CN of forest biomass.

The last step of the scoping review is *summarizing and reporting the findings*. During this step the main findings are presented. Our scoping review revealed that there is no commonly accepted CN definition. Instead, we were able to distinguish eight distinct concepts of CN. Additionally, we identified and analyzed the core parameters that underpin the foundation of CN concepts. The more detailed description of the scoping review methodology used for the analysis of the CN concept as well as the results can be found in Chapter 3.

2.2. Life cycle assessment of residential heat production from wood pellet combustion in the Northwest region of Russia

This section of the thesis briefly describes the methodology used in the publication presented in Chapter 4 of the thesis.

LCA is a method used to analyze the environmental impact associated with a product/system throughout its life cycle. The study was carried out in compliance with the International Organization for Standardization (ISO) 14040 and 14044 guidelines (ISO 2006a, ISO 2006b). OpenLCA software version 2.1.1 (openlca.org) and Ecoinvent 3.4 (Ecoinvent 2017) database were used to conduct current LCA.

LCA methodology consists of several steps: 1) goal and scope definition; 2) life cycle inventory (LCI); 3) life cycle impact assessment (LCIA); and 4) interpretation of the results.

Goal and scope definition

This step of LCA requires establishing the purpose of the study and outlining the system boundaries. In our LCA, the main goal was to analyze the environmental impact of thermal energy generation on a domestic level from combustion of wood pellets produced locally from sawmill residues in the Arkhangelsk area of the Northwest region of Russia. We used a gate-to-grave approach with system boundaries that included the following steps: 1) sawmill residue collection; 2) pellet production; 3) pellet transportation; and 4) heat production and waste disposal. 1 MJ of heat generation was selected as a functional unit.

Life cycle inventory

LCI involves collecting data on all inputs and outputs throughout the life cycle of the product. The inventory data was provided by Executive Director of the “Russian Pellet Union” Association and completed with the data from the Ecoinvent database. The heat energy production scenario in our LCA began with the collection of sawmill residues in the form of wood chips, shavings, and sawdust. Since these residues are a by-product of the lumber industry, it was not necessary to consider forest establishment and forest harvesting processes in our study. After biomass was collected, it was used to produce wood pellets. Pellets were produced directly at the sawmill. The pellet production process consisted of grinding, milling, drying, pelletization, cooling, packaging, and storage. During the drying process, part of the biomass was used as a fuel to produce heat. It should be noted that biogenic emissions from the combustion of biomass were not accounted for in our LCA. If not used for bioenergy production, this biomass would be disposed of via burning or landfilling, therefore emitting the same amount of carbon during combustion or decomposition, respectively. When pellets were produced, it was assumed that they were purchased directly from the producer and transported

to the end user via a light truck over a distance of 20 km. After pellets were transported, they were combusted in a 25kW pellet furnace for heat generation. Combustion waste in the form of ashes was disposed of at a landfill.

Life cycle impact assessment

The LCIA stage consists of evaluation of the environmental impact. We used the ReCiPe 2016 (H) Midpoint v. 1.1 method (Huijbregts et al. 2017) during this step. The environmental impact was assessed across the following categories: global warming (GW), freshwater eutrophication (FE), marine eutrophication (ME), terrestrial acidification (TA), fine particulate matter formation (FPMF), fossil resource scarcity (FRS), ozone formation (human health; OFHH), ozone formation (terrestrial ecosystems; OFTE).

Interpretation of the result

This is the final step of the LCA where the results of the impact assessment were analyzed, and the overall conclusions and recommendations were drawn. Additionally, the robustness of the results was verified by conducting three sensitivity analyses. The first sensitivity analysis used the ILCD 2011 impact evaluation approach (European Commission, Joint Research Centre, and Institute for Environment and Sustainability 2011) as an alternative to the ReCiPe methodology in order to assess methodological influence on the findings across similar impact assessment categories. The second sensitivity analysis evaluated the environmental impacts of transportation distance variations in the pellet distribution stage. The third sensitivity analysis assessed the alternative heating source used for drying during the pellet production process.

A more detailed description of the methodology utilized during the LCA of heat production from pellets combustion in Russia, as well as the results can be found in Chapter 4.

2.3. Methods for monetary valuation of ecosystem services: a scoping review

This section of the thesis briefly describes the methodology used in the publication presented in Chapter 5 of the thesis.

This publication utilizes the same scoping review methodology presented in section 2.1 of this thesis and follows the same guidelines. Likewise, it consists of five main steps: 1) identifying research questions; 2) identifying relevant studies; 3) selecting eligible studies; 4) charting the data; and 5) summarizing and reporting the results.

During the initial step of the review, the research question was formulated: What methods can be used for the monetary valuation of ecosystem services?

The second stage of the study included a literature search in the WoS and Scopus databases using the following key phrases: “ecosystem services assessment method” and “ecosystem services valuation method.” The inclusion criteria were developed consisting of the following requirements:

- Peer-reviewed articles (including case studies) and gray literature
- Written in English
- Any country of publication
- No limit based on the ecosystem type
- Must include a description of method(s) used for monetary valuation of ES

During the third step of the review, the study selection process was conducted based on the inclusion criteria. The flow chart of the study selection process is presented in **Figure 2**. Out of the 109 articles that were initially identified, 16 studies were selected to be included in the scoping review.

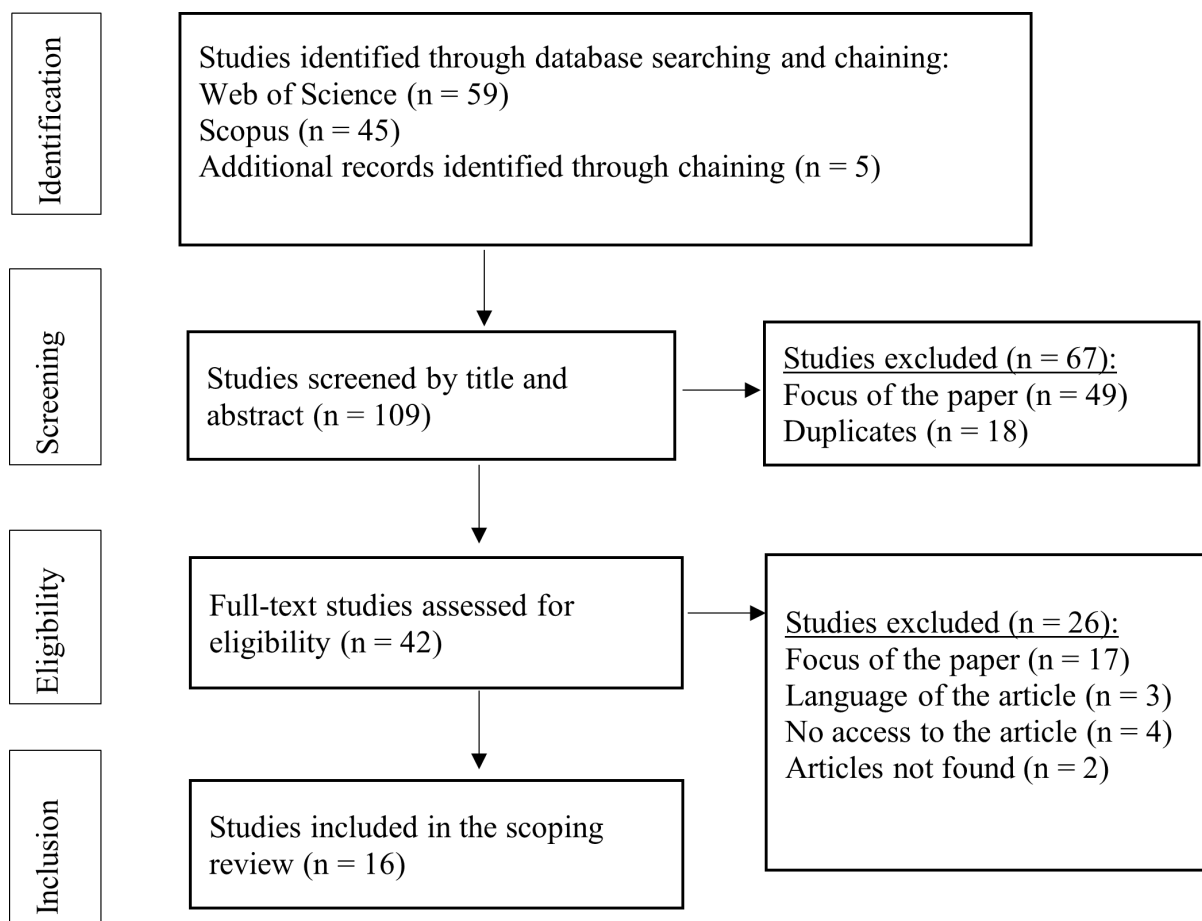


Figure 2. Flow chart of the study selection process of this scoping review

During the fourth step of the review, a charting table was created to facilitate the analysis of the selected articles. This table included information regarding author, year of publication, type of publication, goal of the study, definition of the ‘value’, name of the method for ES valuation, description of the method, and other notes.

The fifth stage of the study focuses on summarizing and reporting the results. During the scoping review of the methods for monetary valuation of ES, we were able to identify over 20 valuation approaches. A more detailed description of the scoping review methodology used for the analysis of methods used for monetary valuation of ES as well as the results can be found in Chapter 5.

2.4. Is forest biomass used for bioenergy production a viable climate change mitigation option?

This section of the thesis briefly describes the methodology used in the publication presented in Chapter 6 of the thesis.

This publication was written in the form of a short communication based on the qualitative synthesis approach that includes a short description of the articles presented in Chapters 3 and 4. The aim of this article is to analyze the concept of CN and its application to the bioenergy production process using the LCA methodology. This article is structured as follows:

Introduction – discusses the potential role of forest biomass in CC mitigation and introduces the concept of CN.

CN section – describes the main CN concepts that were identified during the scoping review (Chapter 3) as well as the key assumption underpinning these concepts.

LCA section – outlines the LCA of heat generation from pellet combustion (Chapter 4) that provides the overview of the LCA methodology and highlights the main results.

Conclusion - discusses key findings from both studies in relation to the research objectives outlined in the introduction section of the short communication article. Additionally, it provides policy recommendations and identifies future research directions.

Chapter 3: Carbon neutrality of forest biomass for bioenergy: a scoping review

3.1. Summary and contribution statement

Citation of the article: Selivanov E, Cudlín P, Horáček P (2023). Carbon neutrality of forest biomass for bioenergy: a scoping review. *iForest* 16: 70-77.

DOI: <https://doi.org/10.3832/ifor4160-015>

As a lead author, my contribution to the scoping review on the carbon neutrality concept included: design of the overall framework of the study; development of the search strategy of relevant literature; screening and selecting of the studies; extraction and analysis of the data; and interpretation of the findings. Additionally, I wrote the manuscript, created tables and figures, and handled each step of the submission process of the publication.

The main goal of this article was to distinguish the most widely used definitions of CN as well as to identify the main aspects that underpin CN concepts.

The main findings of our scoping review showed that there is no commonly accepted definition of CN of biomass. During the literature analysis, we were able to identify eight distinct concepts of CN: carbon cycle neutrality, inherent carbon neutrality, offset carbon neutrality, carbon neutrality of forest residues and waste, carbon neutrality of additional biomass, accounting carbon neutrality, substitution carbon neutrality, and life cycle carbon neutrality. The identified CN concepts provided varying rationales for labeling biomass used for bioenergy as CN. However, during our review, we critically examined the key assumptions underlying CN concepts that can significantly influence the CN status of biomass. These assumptions include temporal and spatial boundaries, the origin of feedstock, and scenario-based assumptions. In conclusion, our findings demonstrate that CN is not an inherent property of biomass and that

the LCA approach should be used to assess the environmental impact of bioenergy production throughout its life cycle.

3.2. Content of the article “Carbon neutrality of forest biomass for bioenergy: a scoping review”

Introduction

The phenomenon of global climate change (CC) is widely accepted in the scientific community. The evidence demonstrating the progression of climate change is comprehensive. Green House Gas (GHG) emission, which occurs due to human activities, is a significant factor contributing to CC (IPCC, 2022a). The gas making up the greatest share of GHGs emitted by human activities is CO₂. According to data obtained in 2020, the amount of CO₂ emitted is 32018 million tons (UNFCCC 2020, BP 2021) and as of February 2022, the mean concentration of CO₂ in the atmosphere has reached 417.81 parts per million (ppm) (NOAA, 2022). Scientists do not dispute the anthropogenic nature of increased concentrations of atmospheric carbon dioxide. Intergovernmental Panel on Climate Change (IPCC) acknowledges that “human influence on the climate system is clear, and recent anthropogenic emissions of greenhouse gases are the highest in history” (IPCC, 2014).

Climate change is a global problem, hence the solutions for this problem should be achieved through common efforts. For this purpose, many international scientific conferences and political conventions are focused on establishing international cooperation and drafting agreements dedicated to climate change mitigation. One example of such cooperation is the Intergovernmental Panel on Climate Change - United Nations body for assessing the science related to climate change, established by the World Meteorological Organization and the United Nations Environment Programme. Moreover, there is a number of international agreements dedicated to developing the policy mechanism for climate change mitigation such as the United Nations Framework Convention on Climate Change (UNFCCC), the Kyoto Protocol and the

most recent Paris Agreement. The latter has a main goal of keeping global average temperature increase well below 2 °C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5 °C.

The European Council has developed several targets for climate change mitigation that should have been achieved by 2020. These targets included reducing GHG emissions to at least 20% below 1990 levels, improving energy efficiency to achieve a 20% reduction in primary energy use compared with projected levels, and increasing the share of renewable energy sources to 20% of EU final energy production (JRC 2014).

Bioenergy is considered to be a part of the solution portfolio to combat climate change. Energy derived from biomass is the largest renewable energy source utilized to meet the above-mentioned targets in Europe (Berndes et al. 2016). A large share of bioenergy is derived from forest biomass. A popular point of view was that forest biomass was a perfect substitute for fossil fuels due to its renewability and carbon neutrality. However, recently the perception of carbon neutrality has been questioned within the scientific community (Norton et al. 2019). In the past decade, many studies were published regarding the issue of carbon neutrality of forest biomass and the debate over this topic is ongoing.

The current study investigates the problem of carbon neutrality of forest biomass and contributes to the developing discussion. This research is conducted in the form of a scoping review. The objectives of this scoping review are to distinguish the most widely used definition of carbon neutrality, and to identify the most frequently discussed aspects of the carbon neutrality concept.

Methods

This study uses the scoping review approach. This approach describes existing literature and includes findings from a range of different study designs and methods. A scoping review is

suitable for identifying and analyzing knowledge gaps, for clarifying general concepts and definitions in the literature, and for identifying key topics within a field of research (Munn et al. 2018). In this scoping review, we used a framework that was originally introduced by Arksey and O'Malley (2005), recommendations from Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews (Tricco et al. 2018), and guidance for scoping review from Joanna Briggs Institute (Peters et al. 2015). The scoping review process used in this study includes five stages: (1) identifying research questions, (2) identifying potentially relevant studies, (3) selecting eligible studies, (4) charting the data, (5) and summarizing and reporting findings.

Identifying research questions and objectives

The objectives of this study are to present the most widely used definition of carbon neutrality and the most frequently discussed aspects of the carbon neutrality concept. The research questions we developed are:

- What is the definition of carbon neutrality?
- What are the main assumptions that carbon neutrality concepts are based on?

Identifying potentially relevant studies

In general, the process of identification of relevant studies allows the researcher to cast a wide net for resource types such as electronic databases, peer-reviewed scientific articles, gray literature (e.g., regulatory data), working papers, etc. The databases chosen for this scoping review were Web of Science (WoS) and Scopus. The time frame of publication ranged from 1990 to July of 2020, with the time frame ending at the time the review was performed. The search strategy for both databases used the following key phrases: “carbon neutrality”, “carbon neutral biomass”, “carbon neutral bioenergy”, and “forest carbon neutrality”. The selected terms could be found in the title, the abstract, or the keywords. Furthermore, a forward-

backward article search (also known as citation chaining) was applied. This search method collects references that are frequently cited in topical papers (Robinne et al. 2020). The forward-backward search method was implemented during the process of study selection. In addition, several articles were obtained through direct contact with authors who are focusing on the study of carbon neutrality of forest biomass.

For the study selection process of this scoping review, we developed a list of inclusion criteria:

- We analyzed both peer-reviewed articles (articles published in peer-reviewed journals) and gray literature (e.g., working papers, reports, etc.)
- All publications must be written in English
- We didn't implement any restriction regarding the country of publication
- All publications must be focused on the concept of carbon neutrality of forest biomass

Selecting eligible studies

Figure 3 summarizes the process of study selection. During the initial search using the chosen terms, we identified a total of 5699 articles – 2630 in Web of Science database and 3069 in Scopus database. During the screening process, we excluded 5463 articles that were not relevant to the topic based on the contents of the title and the abstract and 138 articles were excluded as duplicates. Thus, 98 articles were identified from WoS and Scopus databases and 40 articles were obtained through chaining and direct contact with authors. Finally, we performed a full-text analysis on these 138 articles. 99 studies were excluded after analysis. 87 studies didn't have a focus that correspond with the objectives of this scoping review, 5 studies were not written in English, 1 study was not available in a digital format, and 6 studies were not accessible. Therefore, the final number of articles included in this scoping review is 39.

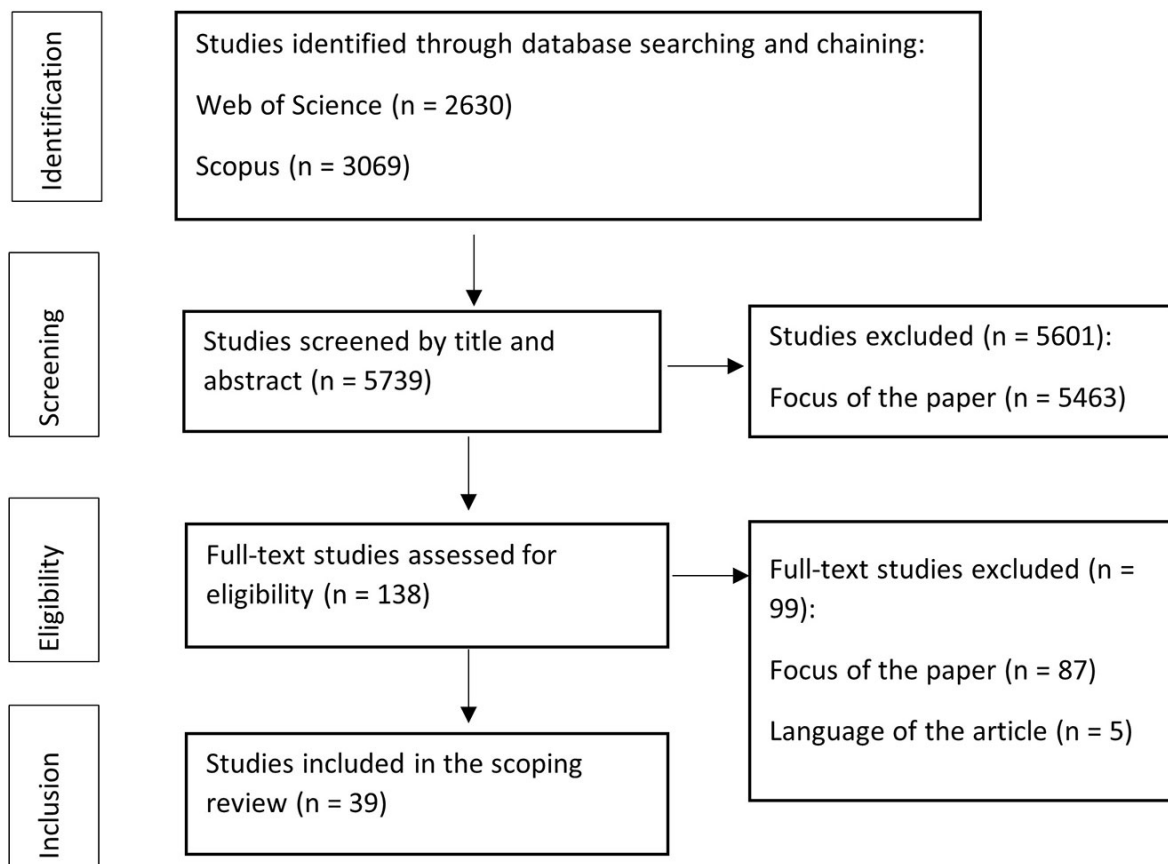


Figure 3. Flow chart of the search strategy

Charting the data

We used descriptive qualitative content analysis and developed a data-charting table in Microsoft Excel for full-text analysis of eligible articles (n=39). This table includes the following information about analyzed articles: author, year of publication, type of publication, goal of the study, definition (concept) of carbon neutrality used in the article, comparison of fossil fuels to biomass with the focus on carbon neutrality, dimensions related to the concept of carbon neutrality that were considered in the article, and general perception of carbon neutrality of forest biomass.

Summarizing and reporting findings

The last stage is summarizing the data in relation to the purpose of the review, making conclusions, and noting any implications of the findings. This stage is thoroughly elaborated

below in the results and discussion sections of the article. The results section explains the CN concepts that were identified during the literature review, and the discussion part of the study explains the assumptions that stay behind the CN concepts.

Results

Study context

39 articles were included in the scoping review after full-text analysis (Table S1). Although inclusion criteria had a time frame from 1990 to 2020, the studies included in the scoping review were published starting from 2007 (Figure 4). 15% of analyzed articles were published between 2007 and 2010, 46% of articles were published between 2011 and 2015, and 39% of articles were published between 2016 and 2020. 16 articles included in the scoping review were published in peer-reviewed journals, whereas 23 articles can be defined as gray literature.

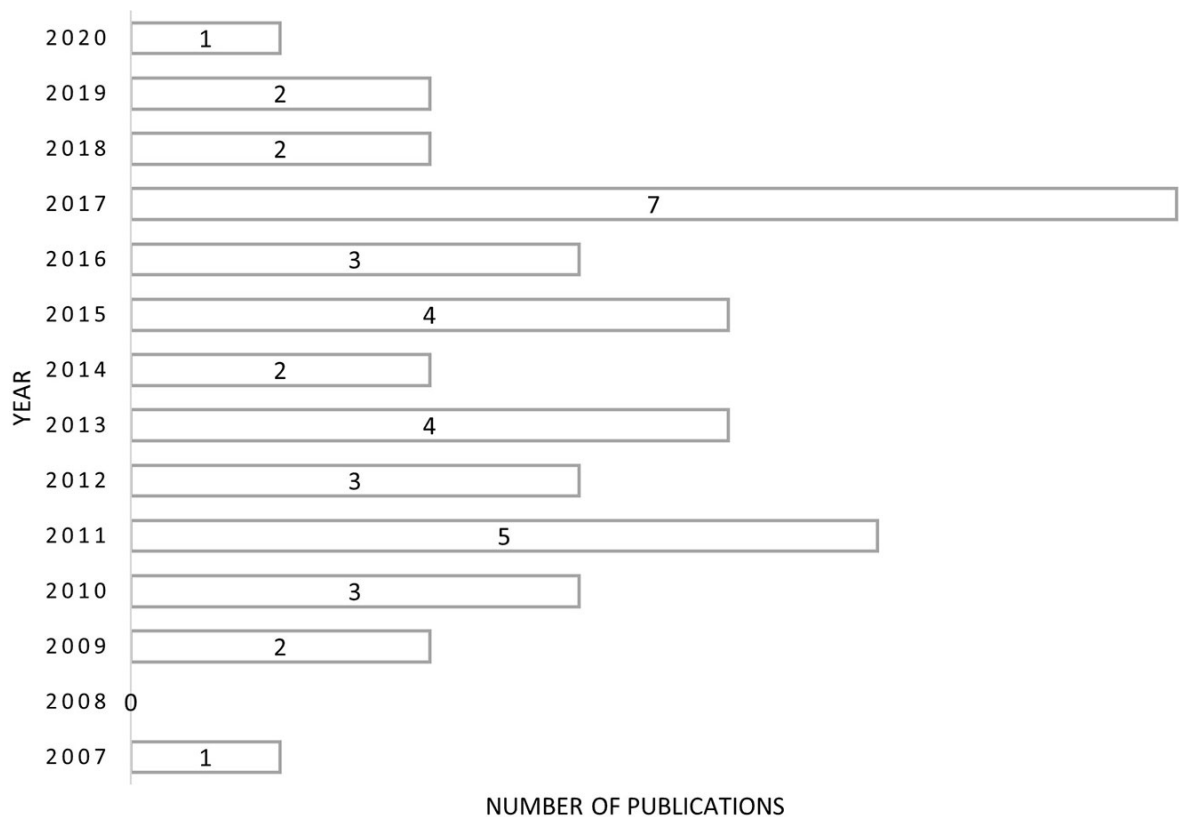


Figure 4. Number of studies included in the scoping review, by year of publication

Concepts of carbon neutrality

The concept of carbon neutrality (CN) of biomass is broad and often vague. Various scientists use this term differently according to their own understanding. As has been pointed out in many studies, there is no widely accepted definition of “carbon neutrality” (NCASI 2013, Bracmort 2016, Liu et al. 2018). In this scoping review, we analyzed which definitions of carbon neutrality are used and which carbon neutrality concepts deem biomass as carbon neutral. After analysis of 39 articles (Table S1), we distinguished 8 different concepts of carbon neutrality. A significant foundation of this research is the classification of carbon neutrality introduced by Malmsheimer et al., (2011) which is based on the materials first presented by the National Council for Air and Stream Improvement (NCASI, 2011). However, we were able to contribute to the discussion on the carbon neutrality of biomass exploitation for bioenergy by ascertaining and defining concepts of carbon neutrality that were not previously formulated (CN of forest residues and waste, and CN of additional biomass). Below are the descriptions of each concept.

Carbon cycle neutrality. According to this concept, forest biomass that was harvested and subsequently replanted can be called carbon neutral (FSG 2015, Nabuurs et al. 2017). Carbon cycle neutrality claims that the same amount of C that was emitted to the atmosphere can be reabsorbed during the process of regrowth after a certain amount of time. Therefore, C simply circulates between atmosphere and biosphere. C released to the atmosphere during the process of biomass combustion creates carbon debt. The process of C sequestration back into plant tissues during forest growth is called the payback period, which can take a few years for short rotation plantations to decades or hundreds of years for old-grown forests. Thus, the concept can be characterized as “harvest to regrowth.” C cycle neutrality is usually applicable at the stand level, meaning that biomass is harvested from one forest plot and regrowth occurs on the same plot. Other forest management activities existing in the rest of the forest are not taken into account.

Inherent carbon neutrality. This concept asserts that biomass is naturally carbon neutral because its combustion returns C to the atmosphere that was recently absorbed during the process of plant growth (Sedjo 2011, Hektor et al. 2016). Unlike fossil fuels which emit C that was sequestered deep in the earth for millions of years, biomass combustion does not add net C to the atmosphere. Inherent CN concept suggests that biomass establishes a C credit during the process of plant growth. When biomass is burned, C is released back to the atmosphere and creates a balance between absorbed and emitted carbon. This concept can be characterized as “growth to harvest,” the opposite of the perspective of carbon cycle neutrality. Yet both inherent CN and carbon cycle neutrality state that biomass C circulates in the closed biosphere-atmosphere system without adding net additional C.

Offset carbon neutrality. The amount of C released due to harvesting in a forest stand can be offset by C sequestration in other stands within the same forest (Sedjo and Tian 2012, Timmons et al. 2016). Unlike carbon cycle neutrality, the offset carbon neutrality concept considers carbon stock of the forest as a system where one stand should not be analyzed in isolation from other stands. If the annual amount of C released from harvesting activities in the forest is less than or equal to the annual amount of C sequestered during the process of forest growth, there is no net reduction in C stock. Therefore, biomass from this forest can be called carbon neutral.

Carbon neutrality of forest residues and waste. Biomass can be considered carbon neutral if the C released during the process of combustion would otherwise be released into the atmosphere via the process of natural decomposition (Vanhala et al. 2013, Cowie et al. 2017). This concept can be applicable for forest residues that were left in the forest after any harvesting activities including clearcuts, thinnings, harvesting to reduce fire risk and risk of pest diseases in the

forest, and harvesting activities caused by construction works. In addition, the concept of carbon neutrality of forest residues and waste can be relevant to sawmill or papermill wooden waste.

Carbon neutrality of additional biomass. Biomass that is planted on non-forested land and is grown explicitly with a purpose to be used as a bioenergy source can be called carbon neutral (Zanchi et al. 2010, JRC 2014). If this biomass were not planted, the C it absorbs would have remained in the atmosphere. This principle is similar to “growth to harvest” and the inherent CN concepts. Biomass creates a C credit during growth that will be balanced to zero after biomass combustion without adding net C emissions to the atmosphere.

Accounting carbon neutrality. This concept is derived from the United Nations Framework Convention on Climate Change (UNFCCC) and Kyoto Protocol principles for international greenhouse gas reporting and accounting. According to these principles, C released during the process of biomass combustion should be counted as zero. The emissions are instead reported in the Land Use, Land Use Change and Forestry (LUCUCF) category at the stage of harvesting as a loss of C stock in the forest. Thus, emissions of C from tailpipes and smokestacks associated with biomass combustion for energy are not included in the emissions from the energy sector to avoid double-counting. Comparable to offset CN, this approach accounts for annual changes in C stocks but not C flows.

Substitution carbon neutrality. In the analysis of 39 articles, the concept of substitution carbon neutrality was rarely defined as an independent concept. However, its principles are relevant to many other CN concepts. Within the framework of this concept, biomass can be considered carbon neutral if the net emissions of C released during the process of combustion of biomass for energy are less than C emissions associated with the exploitation of alternative energy sources (Malmsheimer et al. 2011, Bracmort 2016). Normally, this concept is not limited to

quantifying emissions that occur directly from combustion. When biomass is compared to alternative energy sources (e.g., other renewable energy sources or fossil fuels), it is common practice to use methods of Life Cycle Assessment (LCA). The International Organization for Standardization (ISO 2006a) defines LCA as the “compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product system throughout its life cycle.” A distinctive feature of LCA when applied to the comparison of energy sources is its assessment of all C emissions released over the life cycle, including throughout the steps of production. For example, LCA of biomass will assess the total amount of C released at the stages of harvesting, transportation, processing, conversion, and combustion.

Life cycle carbon neutrality. According to this concept, biomass can be called carbon neutral if it does not release net C emissions to the atmosphere during its life cycle (NCASI 2013, Littell 2016). In other words, the C emissions of biomass should be less than or equal to the amount of C sequestered in the biomass. This concept highlights the necessity of using LCA to assess life cycle emissions of the biomass. Unlike substitution CN, the life cycle CN concept assesses the balance between C released during combustion and the C sequestered during the process of biomass growth without a required comparison of biomass to alternative energy sources.

During full-text analyses of the articles included in this scoping review, we observed that the most frequently mentioned concepts of carbon neutrality are “carbon cycle neutrality” and “offset carbon neutrality”. How often the different carbon neutrality concepts were mentioned in the analyzed articles is shown on Figure 5. It is important to note that we included concepts specifically mentioned as related to carbon neutrality as well as those that were only described without being identified as carbon neutrality by the authors. The average number of carbon neutrality concepts mentioned per article is 3-4.

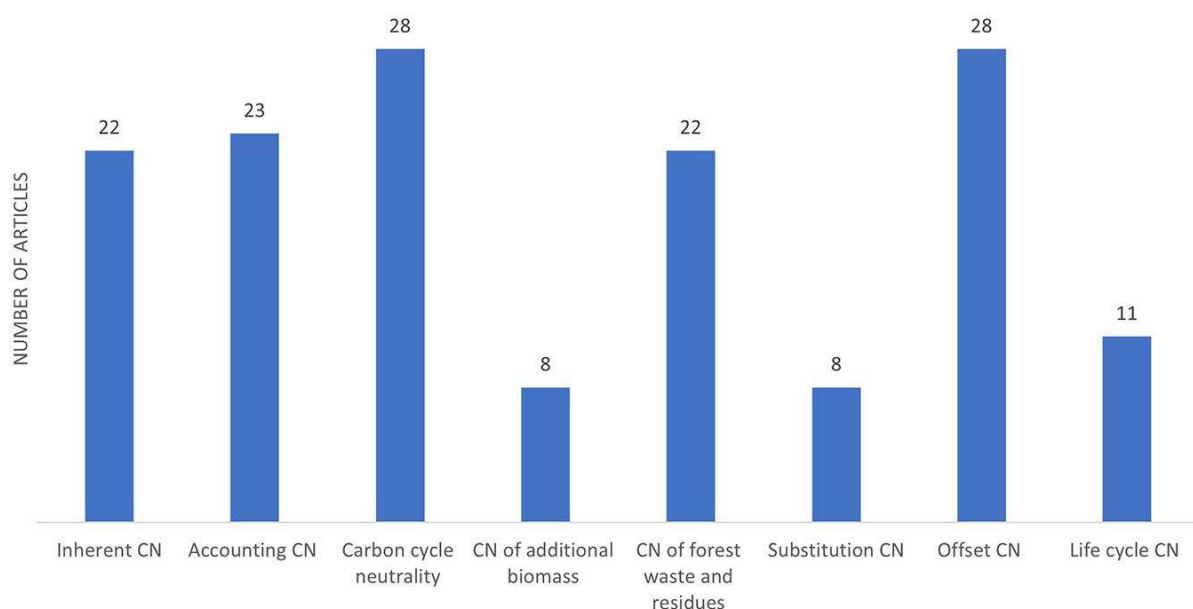


Figure 5. Frequency of mention of carbon neutrality concepts in the analyzed articles

Discussion

Despite the fact that there are numerous concepts of carbon neutrality that could facilitate the use of forest biomass for bioenergy, the overarching idea as well as each individual concept have received a considerable amount of criticism. A key factor that can affect CN of biomass is the method used for biomass assessment. In the analysis of articles, we observed that many authors suggest using LCA to assess carbon neutrality of forest biomass. Currently, the LCA method is widely recognized and applied to bioenergy assessment (Cherubini & Stromman 2011, Wolf et al. 2015, Martin-Gamboa et al. 2020, Musule et al. 2021). Almost all CN concepts (except accounting carbon neutrality) require an assessment of C emissions that can apply this method. The LCA method is not inherently a carbon neutrality concept but is an essential tool to evaluate emissions of a biofuel during its life cycle. Life cycle emissions include supply chain emissions (harvesting, transportation, processing) and emissions from combustion. LCA uses different input parameters to estimate C emissions, and the carbon neutrality of biomass will depend on the parameters used for the analysis (Helin et al. 2013). These parameters include the type of biomass feedstock, aspects of biomass management and harvesting, means

of transportation, the methods of biomass conversion and combustion, temporal and spatial parameters, and the time frame for feedstock replenishment. Some CN concepts highlight a single parameter while others focus on a combination of these parameters. For example, carbon neutrality of forest residues and waste is focused on the origin of the biomass feedstock. Carbon cycle neutrality highlights feedstock replenishment, whereas inherent carbon neutrality prioritizes the temporal aspect of biomass production. Other CN concepts focus on a combination of these parameters; for instance, the offset CN concept considers both temporal and spatial boundaries, and additional biomass CN distinguishes between origins of biomass feedstock as well as considers the temporal aspect. However, life cycle carbon neutrality concept is unique in its reliance on the LCA method and has the most thorough assessment of all parameters that can affect the CN status of a biofuel. The concept of substitution CN can also implement the LCA method to assess how bioenergy production compares to alternative energy production scenarios in terms of C emissions.

Temporal boundaries

While assessing C emissions produced during the entire life cycle of biomass, one of the challenges is to properly define the time frame of the analysis. This element of analysis is the establishment of “temporal boundaries”. Depending on the length of the designated time boundaries, biomass would be characterized as carbon neutral when C emissions from bioenergy production will be less than C sequestration in a selected time frame. It can be achieved if the boundaries will include C reabsorption *via* biomass regrowth after harvesting for bioenergy or C sequestration during the initial biomass growth. This parameter is relevant for concepts of carbon neutrality including carbon cycle CN, life cycle CN, inherent CN, and CN of additional biomass. Thirty-four out of 39 articles analyzed during the scoping review discussed at least one of these CN concepts, thereby considering temporal aspects. There are many opinions about when the accounting period should begin since this choice can influence the perception of biomass as CN and there is no agreement on which temporal boundaries

should be applied across the board. The debate is ongoing as scientists provide individual explanations for the ideal temporal boundaries with which to determine carbon neutrality. One option is to begin accounting at the time of first biomass harvesting for bioenergy (Berndes et al. 2016). In this case, biomass extraction creates C debt that later can be repaid *via* replanting of biomass. Another option is for accounting to start at the point of biomass planting and growth (relevant for inherent CN and CN of additional biomass) where biomass initially creates C credit.

It is just as important to determine when the accounting period should end. One option is to account for growth prior to harvesting, harvesting activities, and full biomass regrowth (Littell 2016). A relatively long time frame is suggested because forest management activities commonly occur before and continue after the usual timeframe of LCA analysis. On the one hand, longer temporal boundaries will allow for the carbon debt that occurs at the moment of harvesting to be repaid by regrowth and describe the GHG mitigation benefits provided by stable forest carbon stocks (FSG 2015). On the other hand, it is argued that longer temporal boundaries should not be applied when assessing CN of biomass because they overshoot established Paris Agreement targets and hinder efforts to mitigate climate change (Diehl et al. 2020). In many cases, the actual carbon payback period is very long (Holtmark 2012). When biomass is combusted, it immediately increases the concentration of C in the atmosphere, and it is theorized that a delay between C release and C sequestration could have a temporary warming effect (Muys et al. 2014). Some scientists argue that burning biomass with a long payback period has a potential impact on climate tipping points in the short term (Agostini et al. 2017, Brack 2017a). The concept of “climate tipping points” implies that an increase in global temperatures past a certain point could potentially cause an irreversible global shift from the current stable climate to a warmer stable climate and many subsequent negative consequences (Brack 2017a). For example, thawed permafrost and melted glaciers would not return to their original magnitude at a higher temperature equilibrium even after full carbon

emissions offset by biomass regeneration. Nevertheless, this issue has not yet been sufficiently studied, and research is ongoing.

Spatial boundaries

While assessing the emissions released from biomass, it is also important to correctly identify the physical border of the study area. This aspect of assessment is referred to as spatial boundaries. Just like temporal boundaries, spatial boundaries can influence the calculation of biomass emissions (Muys et al. 2014). Therefore, whether or not an examined biomass type is carbon neutral depends on selected spatial boundaries. This aspect is the foundation of offset carbon neutrality. There are two common options for spatial boundaries: plot-level approach and landscape-level approach. The plot-level approach assesses a single plot with harvesting at year 0 followed by subsequent regrowth. In that case, the amount of time required for regrowth (payback period for C sequestration) will be significant. The landscape-level approach considers an entire forest area. According to this approach, emissions from harvesting on one plot can be offset by forest growth in other plots in the same forest. Therefore, biomass from this forest can be considered carbon neutral if the total carbon stock remains the same or increasing despite the undertaken harvesting activities. The more widely held opinion is that the landscape-level approach is the most suitable for assessing the C emissions of biomass (Malmsheimer et al. 2011, Cowie et al. 2017, IEA 2019) and that the minimal spatial boundaries should include the entire supply area (NCASI 2013, FSG 2015). One reason for this constraint is the possibility to account for “leakage”. Leakage can be defined as unexpected emissions or additional sequestration associated with forest management activities within the forest plot but occurring outside of the assessed area (FSG 2015). Plot-level assessment does not consider emissions or sequestration related to unexpected forest activities, whereas the landscape-level approach allows for more thorough measurement within a larger area of consideration.

However, some argue that the C balance in the areas of the forest unaffected by direct forest management activities is irrelevant and that emissions of a specific bioenergy project cannot be

offset by biomass growth in the adjacent area (Diehl et al. 2020). In addition, a report published by the Manomet Center for Conservation Sciences (MCCS 2010) points out that landscape-level assessment ignores the scenario in which biomass was not harvested for bioenergy (*i.e.*, business as usual scenario) because it is hard to predict forest management activities when applying wider spatial boundaries. Considerations in biomass assessment, such as the BAU scenario, and possible omissions of foregone C sequestration will be explained in the next section.

Scenario-based assumptions

As is pointed out by many authors (Haberl et al. 2012, Matthews et al. 2018), it is essential to apply assumptions from different scenarios when assessing GHG emissions of biomass. This component of assessment can be applied to all CN concepts. When assessing biomass, it is reasonable to compare the “bioenergy scenario” – the scenario in which biomass is harvested for energy production through processes of its combustion – with the scenario in which biomass is never harvested for bioenergy production. This scenario can be called baseline, business-as-usual (BAU), reference, or counterfactual scenario. However, BAU scenarios can vary. For example, if a forest has been actively managed with a certain amount of biomass consistently harvested, the BAU scenario assumes that the same amount of biomass will continue to be harvested in the future. Conversely, if forest management activities were absent in the assessed forest area, the BAU scenario assumes that the forest will remain untouched and continue to absorb carbon as it grows. The process of ongoing plant growth in a non-bioenergy scenario is called foregone carbon sequestration. Foregone carbon sequestration poses a challenge to the promotion of forest biomass for bioenergy production and to its categorizing as carbon neutral. Many authors are concerned that encouraging the harvest of biomass for energy will lead to the loss of potential carbon stock (EEA 2011, Muys et al. 2014, Agostini et al. 2017, Liu et al. 2018). Additionally, biomass can only reduce C emissions if the total C sequestered after the processes of growth and harvesting of biomass is greater than the amount that would have been

sequestered anyway (Searchinger et al. 2009). While they may be important to consider, counterfactual scenarios are often hard to predict. There are a variety of factors that can impact future forest conditions that may or may not make biomass suitable as a carbon neutral source for energy production. The potential risk of natural disturbances like fires, storms, or pest diseases is often mentioned among these factors (Matthews et al. 2018). Another consideration is market effects. The conditions of the current economic market (*e.g.*, prices of biomass), or even the anticipation of future markets, can affect forest owner behavior. Making decisions regarding forest management activities based on expected future markets can be called a forward-looking approach (Sedjo 2011, 2013). According to this approach, high market prices for biomass, or anticipated high prices in the near future, can increase forest owner investment and improve forest management. Improvement of forest management can mean new plantings, the use of more rapidly growing seedlings, the intensification of management to increase quality and quantity of feedstock, and expansion of forests (Sedjo 2013, Ter-Mikaelian et al. 2015, Cowie et al. 2017, Nabuurs et al. 2017). In that case, these positive effects should be included in the BAU scenario. However, market effects can also have negative impacts on the forest. High prices of timber, as well as large subsidies for bioenergy, can increase the rate of harvesting. If harvestings are not conducted within the sustainable forest management framework, the C stock can eventually be diminished (Sedjo 2013, Diehl et al. 2020). Forest owner behavior, market effects, and natural ecosystem dynamics are difficult to forecast. Consequently, the fate of C stocks is difficult to predict as it is closely associated with these factors. This demonstrates why assessments of the exploitation of biomass for bioenergy should precisely define alternative scenarios and include all the details that can affect the results of the assessment.

Many authors assert that it is necessary to account for supply chain emissions when assessing emissions from biomass (Cowie et al. 2017, Liu et al. 2018, Norton et al. 2019). Even though the energy content of the fossil fuels used for harvesting, chipping and transportation is typically

less than 5% of the energy content of the processed biomass (Berndes et al. 2016), emissions related to the combustion of these fossil fuels should be included in LCA. According to this logic, a comprehensive comparison between a bioenergy scenario and a counterfactual scenario in which only fossil fuels are used should consider the C sequestration of the biomass that continues to grow in the counterfactual scenario as well as the supply chain emissions from fossil fuels used for mining (or harvesting), transportation, and production in either scenario. This idea is relevant to the substitution CN concept.

Feedstock

The feedstock of biomass for bioenergy is an important factor examined in the analyzed articles. Selection of biomass feedstock affects the overall suitability of its use as an energy source. Currently, the main feedstocks used for bioenergy consist of byproducts from the production of pulp, paper, and sawn timber, as well as from forest residues left after various forest management activities (Zanchi et al. 2012). Forest residues and waste seem to be favorable biomass feedstocks that can positively contribute toward climate change mitigation (Brack 2017b, Cowie et al. 2017, Norton et al. 2019). According to the principle of CN of forest waste and residues, this biomass can be defined as C neutral if used for bioenergy because it would decay and emit C into the atmosphere if it had been left unused.

A related dilemma is whether it is appropriate to harvest biomass in old-growth forests or if it is better to maintain it as a carbon pool. There are two interconnected aspects regarding this issue. First, some authors assert that forest C sequestration capacity is limited with age (Hektor et al. 2016, Timmons et al. 2016). Rates of carbon uptake decline as forests reach maturity. Whereas young forests grow rapidly and sequester higher amounts of C, old-growth forests eventually stop absorbing additional C and only function as C storage. Second, old-growth forests have a higher risk of natural disturbances. There is general agreement that the risk of forest fires, storm fellings, and insect attacks increases with the age of the trees (Malmsheimer

et al. 2011, Hektor et al. 2016). The loss of C pools in an old-growth forest as a result of natural disturbances would be significantly greater than if the disturbance occurred in a young forest.

The link between climate change and forest fires (Abatzoglou & Williams 2016, Flannigan et al. 2006) has increased the frequency and severity of wildfires in recent years and has led to the degradation of both old-growth and young forest areas. As of October 10th, 2022, land burned by wildfires in Europe was estimated at 1.4 million hectares. This is a significant increase considering about 0.45 million ha of land were burned in Europe on average between 2006 and 2021 (Statista 2022). It is recognized that high tree stand density correlates with high fire severity (Amato et al. 2013). Accordingly, forest management practices can positively affect wildfire mitigation. Research shows that forest fuel treatments (*i.e.*, thinning and prescribed fire) can decrease fire severity, crown fire occurrence, and lower C emissions related to wildfire (Reinhardt & Holsinger 2010). Moreover, Madrigal et al. (2017) demonstrated that collection of forest biomass for energy production reduces fire risk if forest management activities are done at a landscape level and if biomass is derived from thinning and understory clearing. Thus, proper forest management facilitates wildfire mitigation while providing biomass resource for bioenergy production.

Conclusion

Despite the wide use of the term carbon neutrality, an analysis of recent literature demonstrated the lack of a common definition of the concept. We were able to identify eight distinct definitions of CN from 39 analyzed articles. These definitions include the following carbon neutrality concepts: carbon cycle neutrality, Inherent carbon neutrality, offset carbon neutrality, carbon neutrality of forest residues and waste, carbon neutrality of additional biomass, accounting carbon neutrality, substitution carbon neutrality, and life cycle carbon neutrality. The majority of investigated studies used more than one definition of CN. Based on the analysis of the selected articles, CN is not an inherent property of biomass. Meanwhile, the inherent

status as carbon neutral is frequently used to promote and justify the utilization of biomass for bioenergy purposes. Using the term in a generic sense can be confusing and even misleading given the variety of concepts of CN. Therefore, the concept of CN should perhaps instead be understood as a relative characteristic of a bioenergy product based on an assessment of the carbon emissions and environmental impact caused by its production and exploitation. This definition is most correlated with the life cycle carbon neutrality concept. Biomass used for bioenergy production can be labeled as carbon neutral using a clear assessment procedure. This can be accomplished by applying the LCA method. LCA provides a tool to investigate all GHG emissions throughout the biofuel life cycle starting at the biomass plantation phase, following through the processes of biomass preparation and transformation, and ending with biofuel combustion. Due to the context-reliant nature of the concept of carbon neutrality of biomass, the results of the LCA scenario that utilizes biomass for bioenergy purposes should be compared to the results of an alternative scenario in which conventional fossil fuels are used as the source of energy production. Furthermore, it should be noted that the LCA of a particular biomass can be affected by systemic, temporal, or spatial boundaries. Therefore, these boundaries should be thoroughly defined before finalizing any assessment. A topical example of biomass assessment is within the framework of RED II. The term “carbon neutrality” is not explicitly stated in RED II (EC 2018), but the CN concepts used can be identified from the directive’s guidelines for assessing the sustainability of biomass fuels and associated GHG emissions during its life cycle. The directive argues that carbon emissions from biofuel combustion should be treated as zero which indicates the principles of inherent carbon neutrality. Additionally, RED II agrees with the principles of the offset CN concept. The directive recognizes biomass as sustainable and proposes that it can be used for bioenergy production as long as levels of carbon stocks and sinks in the forest are maintained after harvesting. Overall, the methodology recommended for sustainability and GHG emissions assessment provided in RED II correlates with the life cycle CN concept which considers emissions that occur during the biomass cultivation,

transportation, processing, etc. Nevertheless, RED II does not clearly define temporal boundaries which could affect the results of the assessment suggested in the methodology. Temporal boundaries, as well as the feedstock- and scenario-based assumptions mentioned earlier in this article, should be considered by stakeholders and policymakers when discussing bioenergy production.

Supplementary materials

Table S1. CN concepts mentioned in the analyzed articles from scoping review.

Author, year published	Inherent CN	Accounting CN	Carbon cycle neutrality	Offset CN	CN of forest waste and residues	Substitution CN	CN of additional biomass	Life cycle CN
NCASI, 2013	+	+	+	+		+		+
Abbasi, T., Abbasi, S., 2010	+							
Woodworth, 2013	+		+	+	+			+
FSG, 2015	+	+	+	+		+		+
Berndes et al., 2016	+	+		+	+			
Brack, D., 2017b	+	+			+			
Brack, D., 2017a	+	+	+	+	+			+
Cowie et al., 2017	+			+	+			
Agostini et al., 2017		+		+				
EEA, 2011	+	+	+	+	+			
Bracmort K., 2016	+		+		+	+		+
Manomet Center, 2010	+		+	+	+		+	
Muys et al., 2014	+	+						
Strauss, W., 2011			+	+				
Littell, D., 2016	+		+	+	+		+	+
Ter-Mikalian et al., 2015	+		+	+			+	
Sedjo, R., 2011	+		+	+	+	+		

Zanchi et al., 2010		+	+		+	+	+	+
Johnston, C., van Kooten, C., 2015			+	+	+			
Nabuurs et al., 2017			+	+				
Zanchi et al., 2012		+	+	+	+		+	
Hektor et al., 2016	+			+				
Malmsheimer et al., 2011	+	+	+	+		+		+
Miner, R., 2007	+	+		+				
Norton et al., 2019		+	+	+	+			+
EASAC, 2017		+	+	+	+			
IEA, 2019	+	+	+	+		+		
Diehl et al. 2020		+		+		+		
Sedjo R., 2013		+	+	+				
Liu et al., 2018	+	+	+		+			
Searchinger et al., 2009		+			+		+	
Haberl et al., 2012	+	+	+	+	+		+	
JRC et al. 2014		+	+		+		+	+
Johnson E., 2009	+	+						
Sedjo, R., Tian, X., 2012			+	+				
Matthews et al., 2018			+	+	+			
Timmons et al., 2016			+	+	+			
Vanhala et al., 2013		+	+		+			+
Rogeli et al., 2015			+					

Chapter 4: Life cycle assessment of residential heat production from wood pellet combustion in the Northwest region of Russia

4.1. Summary and contribution statement

Citation of the article: Selivanov E., Cudlín P. (2025): Life cycle assessment of residential heat production from wood pellet combustion in the Northwest region of Russia. *Journal of Forest Science* 71(4): 182-194.

DOI: <https://doi.org/10.17221/88/2024-JFS>

As the lead author of this LCA study, my contribution included: acquisition of the software and the database required for the LCA; acquisition of the necessary inventory data; modeling of each step of the bioenergy system; analysis of environmental impacts; conducting sensitivity analyses; interpretation of the results in context of the existing LCA literature. I also wrote the manuscript, created tables and figures, and handled the publication process.

This main goal of this article was to evaluate the environmental impact of domestic heat generation in the Arkhangelsk area in the Northwest region of Russia from combustion of wood pellets produced from sawmill residues.

The LCA approach was used in order to assess the environmental effects from bioenergy production throughout every stage of the life cycle (pellet production, transportation, and combustion) across the following ReCiPe (H) 2016 midpoint impact categories: global warming (GW), freshwater eutrophication (FE), marine eutrophication (ME), terrestrial acidification (TA), fine particulate matter formation (FPMF), fossil resource scarcity (FRS), ozone formation (human health; OFHH), ozone formation (terrestrial ecosystems; OFTE). The results of the study indicate that the pellet combustion stage is the most impactful on the environment, contributing to the greatest number of the impact categories. The pellet production

stage contributes the most to GW, ME, and FRS impact categories, while the transportation stage had the least overall environmental impact.

Additionally, three sensitivity analyses were carried out in order to confirm the robustness of the findings. The first sensitivity analysis utilized an alternative impact assessment method, using the ILCD 2011 approach to compare results obtained utilizing the ReCiPe method across similar impact categories. The findings were consistent, reinforcing the robustness of the study. The second sensitivity analysis examined the use of natural gas (instead of biomass) as an alternative energy source for the drying process during pellet production step, contrasting with the bioenergy scenario presented in our LCA. The results showed increased emissions in GW and FRS impact categories, while ME and OFTE remained unchanged. A marginal reduction in emissions was observed for FE, TA, FPMF, OFHH. However, the overall remission reduction did not exceed 2%, except for OFHH, which showed emission reduction of 8.7%. The third sensitivity analysis tested the assumed transportation distance. The environmental impact from increasing the distance from 20 km (baseline scenario) to 100 km was evaluated. The findings showed significant increase in emissions in all impact categories, with the biggest growth observed in GW and FRS (both 50% growth), and FE, TA, OFHH, and OFTE (approximately 30% growth for these categories).

4.2. Content of the article “Life cycle assessment of residential heat production from wood pellet combustion in the Northwest region of Russia”

Introduction

Increasing global demand for energy production, negative environmental effects from fossil fuels exploitation, and fossil fuel exhaustibility (Shafiee and Toplan 2009) have led to a growing interest in sustainable and environmentally friendly energy solutions. According to the Renewable Energy Directive II (EC 2018), Europe aims to achieve a 32% share of renewable

energy in total energy consumption by the year 2030. Bioenergy is becoming a more frequently utilized alternative to fossil fuels when compared to other renewable energy resources (Mandley et al. 2020). Woody biomass is a unique resource that can be derived in different forms from a variety of sources, and used for the production of heat, electricity, and transportation fuel (Malmsheimer et al. 2008; Guo et al. 2015).

To facilitate the transition from fossil fuels to renewable bioenergy, bioenergy production options should be carefully assessed. Life Cycle Assessment (LCA) is a commonly used tool for evaluating the environmental impact of a product system throughout its entire cycle. In recent years, numerous studies have analyzed the life cycle of various bioenergy production scenarios (Agostini et al. 2020; Martin-Gamboa et al. 2020; Musule et al. 2021). Many aspects of bioenergy production, such as impacts of different biomass feedstock on biofuel production (Sjølie and Solberg 2011; Morrison and Golden 2016), the nature of the biofuels themselves (Alizadeh et al. 2023), conversion technologies (Carotenuto et al. 2022), and transportation options (Pierobon et al. 2015; Vera et al. 2019) have been thoroughly evaluated. Several studies focused on wood chips production and utilization (Fitzpatrick 2016; Hammar et al. 2017; Klavina et al. 2017). Other studies investigated the impact of briquettes (Alanya-Rosenbaum et al. 2018; Medina-Rios et al. 2021; Sahoo et al. 2021).

One of the most studied and utilized biofuels is wood pellets (Musule et al. 2021). Numerous LCA studies have analyzed various scenarios of pellet production and bioenergy generation from pellet utilization. For instance, Pergola et al. (2018) compared the environmental impact of pellet production from roundwood logs to pellets from sawmill residues (mainly sawdust). Another paper by Lu and Hanandeh (2017) studied various biofuel and bioenergy production scenarios using different conversion technologies for woodchips and wood pellets. Ahmadi et al. (2020) investigated the difference between power generation in Canada from gasification of woodchips originating from softwood harvesting residues to gasification of pellets, combustion of pellets, and pyrolysis of woodchips. McKechnie et al. (2011) examined emissions related to

electricity production from co-firing pellets obtained from different sources. Porso & Hansson assessed the environmental impacts of district heat production via combustion of wood pellets produced from poplar and willow. Wang and colleagues (2017) compared emissions from heat production via combustion of wood pellets to heat production via combustion of coal in Changchun City, China.

Due to increased pellet production and their higher combustion and heating efficiency compared to conventional firewood, pellets are becoming more commonly used for household heat production. Several LCA studies have focused on assessing the environmental impact of pellet combustion for residential heat generation. For example, Sgarbossa et al. (2020) evaluated the environmental profile of four wood pellet supply chains for heat production from different types of biomass feedstock. Research performed by Quinteiro et al. (2019) examined the environmental impact of residential heat production in Portugal by comparing alternative maritime pine wood pellets production scenarios. Pa et al. (2013) evaluated emissions related to pellet combustion for residential heat production in British Columbia. Roder and Thornley (2017) investigated the use of waste wood as a bioenergy feedstock for heat production in the UK. A study conducted by Ferreira et al. (2018) compared heat production from the combustion of wood pellets to pellets produced from grape stalks. Another study published by Quinteiro et al. (2020) compared the environmental impact of heat production via combustion of pellets produced at the sawmill to pellets produced at the household level. Sadaghiani et al. (2023) assessed the emissions related to heat generation from pellet combustion in remote communities in Canada and compared it to heat generated from diesel utilization. Topić Božič et al. (2024) conducted an LCA that compared the effects of heat production via combustion of wood pellets vs. firewood at the household level in Slovenia.

Despite being one of the biggest pellet producers in the world, Russia has not been extensively studied in terms of life cycle assessment of pellet production and utilization. According to FAOSTAT (2024), over two million tons of wood pellets were produced in 2022. More recently,

the export of wood pellets has decreased, creating an opportunity for the transition from fossil fuels to the more environmentally friendly option of utilizing the produced and unsold pellets within Russia for energy needs. Closing this gap in scientific literature is important given the amount of pellets produced in the Russian Federation and the recent trends in pellet export from Russia. This study aims to fill that gap. Using primary data on pellet production in the Arkhangelsk area of the Northwest region of Russia, provided by the Executive Director of the “Russian Pellet Union” Association, this LCA evaluates the environmental impact of residential heat production from pellet combustion.

Material and methods

LCA is a widely utilized method for assessing the environmental impact associated with a product by gathering all relevant inputs and outputs of the product system, evaluating all the impacts associated with every step of the life cycle, and interpreting the results according to the defined goal and scope of the LCA. The current study was performed in accordance with ISO 14040 and 14044 standards (ISO 2006a, ISO 2006b). The assessment was performed using OpenLCA software version 2.1.1 (openlca.org) and Ecoinvent 3.4 database (Ecoinvent 2017). OpenLCA software, as well as Ecoinvent database, were selected for this project because they comply with the ISO standards, are widely accepted in the scientific community, have all the necessary features and capabilities to undertake the project, and are affordable and easily accessible. The ReCiPe 2016 (H) Midpoint v. 1.1 (Huijbregts et al. 2017) was used as a method for quantitative life cycle impact assessment (LCIA) across the following impact categories:

- global warming (GW),
- freshwater eutrophication (FE),
- marine eutrophication (ME),
- terrestrial acidification (TA),
- fine particulate matter formation (FPMF),

- fossil resource scarcity (FRS),
- ozone formation (human health; OFHH),
- ozone formation (terrestrial ecosystems; OFTE).

This particular impact assessment method was chosen because it complies with ISO standards of transparency, scientific rigor, and consistency. It provides a comprehensive and harmonized set of midpoint impact assessment categories that are suitable for this LCA, and it is widely accepted in the scientific community.

Goal and scope

The goal of this study is to evaluate the environmental impact of residential heat production in the Northwest region of Russia from combustion of wood pellets locally produced from sawmill residues in a pellet furnace. A gate-to-grave approach of LCA was applied with system boundaries consisting of the following steps: 1) sawmill residue collection at the sawmill; 2) pellet production; 3) packed pellet distribution; and 4) thermal energy production and waste disposal. The production of fuels, lubricants, low-density polyethylene (LDPE) bags, as well as ash disposal from biofuel combustion, were also accounted for. The production of capital goods (buildings, equipment, and machinery) was excluded due to the scarcity of relevant data, the potential to increase uncertainty, and low relevance to the key impact categories that are analyzed in this study (Silva et al. 2018). 1 MJ of thermal energy produced as a result of combustion of pellets was chosen as the functional unit. A number of sensitivity analyses were performed in order to assess the influence of methodological assumptions and inputs on the environmental impact. The first sensitivity analysis investigates an alternative life cycle impact assessment method to compare the general trends in environmental impact categories. The choice of a particular impact assessment method can influence the results and interpretation of an LCA. Therefore, using an alternative impact assessment method helps to avoid methodological bias, ensures the robustness of the result, and facilitates a more comprehensive

understanding of the environmental impacts. The first sensitivity analysis compares the outcomes of bioenergy production evaluated using the ReCiPe 2016 (H) Midpoint impact category method with those obtained from the ILCD 2011 methodology (European Commission, Joint Research Centre, and Institute for Environment and Sustainability 2011). The ILCD 2011 Midpoint method was selected as an alternative due to the similarity in the analyzed impact categories. These two LCIA methods can be compared across six impact categories. Specifically, the GW, FE, ME, TA, FPMF, and FRS impact categories in the ReCiPe LCIA method correlate with the climate change (CC), freshwater eutrophication (FE), marine eutrophication (ME), acidification, particulate matter (PM), and mineral, fossil, and renewable resource depletion (MFRRD) impact categories in the ILCD 2011 Midpoint method. The second sensitivity analysis assesses the impact of different transportation distances of pellet transportation on the environment. This particular parameter was selected based on the existing literature that highlights the significance of emissions related to the transportation distance in bioenergy production scenarios on overall environmental impact (Beagle and Belmontt 2019; Topić Božič et al. 2024). The third sensitivity analysis focuses on an alternative source of heating used during the drying step of the pellet production process. This aspect was chosen for the third sensitivity analysis as it is one of the most analyzed parameters in bioenergy LCA studies (Martin-Gamboa et al. 2020) and the choice of energy source during pelletization stage can significantly affect the sustainability of the bioenergy production process. A more detailed explanation regarding sensitivity analyses is provided in the *Sensitivity analysis* section.

Description of the region

The bioenergy production scenario has been modeled using primary data obtained via a questionnaire from the Executive Director of the “Russian Pellet Union” Association. The “Russian Pellet Union” is an independent, non-governmental association of pellet producers created to provide a platform for pellet producers to solve the current problems of the industry

in the domestic and foreign markets. It is a self-regulated organization that interacts with government agencies and pellet companies, to solve problems regarding production, logistics, domestic consumption, etc. The Union members account for more than 60% of the production and export of wood pellets in Russia. For this study, several representatives of the pellet production industry in Russia were contacted, but only one industry professional agreed to collaborate. The data describes the production of wood pellets from sawmill residues in the Arkhangelsk area of Northwest Russia and the subsequent local thermal energy production via pellet combustion at the household level. The Arkhangelsk area is located in the Northwest part of Russia and occupies 590 thousand km², of which more than 60% is covered with boreal forests dominated by coniferous tree species. The local climate is subarctic with long cold winters and short summers. The average temperature in winter is 11.6°C; the average summer temperature is 13.6°C. The region has roughly one million inhabitants. This region was selected because it is a major hub for forestry and timber processing with several pellet production mills located in the area. We were able to gather the data about one pellet mill that is located in the center of the Arkhangelsk region (in the suburban industrial area approximately 15 km away from the city of Arkhangelsk), but we were asked not to disclose the name of the facility.

Description of the system

Initially, trees (predominantly Scots pine, with smaller shares of spruce, Siberian fir, and larch) are mechanically harvested for lumber production and transported as logs to a timber mill. At the mill, timber processing generates sawmill residues, such as sawdust, wood chips, and wood shavings, which are collected for pellet production. The pellet production process occurs at the same facility where the biomass is collected. Since the biomass used for pellet production is a by-product that would otherwise be disposed of as waste, the scope of this LCA excludes harvesting, log transportation, and sawmill residue production. Therefore, the system boundaries include the following steps: 1) pellet production, 2) pellet transportation, and 3)

pellet combustion and waste disposal (Figure 6). Pellet combustion and waste disposal are combined into a single process as it is supported by the software and aligns with common practices in the literature (Dias et al, 2017; Ruiz et al, 2018; Quinteiro et al, 2019). Combining these processes ensures a more streamlined and accurate representation of the system, as the waste generated (e.g., ash) is directly tied to combustion activity.

The pellet production stage starts with grinding woodchips and larger residues in a grinder. The process is repeated until the desired size of biomass is achieved. Afterwards, the sawdust collected as sawmill residues is combined with the ground biomass produced during the previous stage and then supplied to the drying drum. During drying, the exhaust air is captured along with small fractions of biomass using a vacuum and deposited down into the hammer mill for further fine crushing. It is important to mention that woodchips and larger residues are used as fuel during the drying process. In a hammer mill, biomass is milled to the size of sawdust (4mm length, 1.5 mm diameter). Next, sawdust is supplied to a cyclone in order to separate the dust from the air. After being filtered from the air, the dust is supplied to the straight screw conveyor and into the pelletizer. The pelletization process occurs in the press-granulator under high pressure and at a temperature of 250-300 °C. Wood dust is glued together via adhesive (e.g., maize starch) in the form of cylindric pellets. Then, pellets are cut by a special blade that is installed in the granulator. The size of produced pellets is 10-30 mm in length, and 6-10 mm in diameter. Cut pellets are hot and have to be cooled down. Pellets are put through a cooler equipped with fans. The cooling process changes the properties of the pellets to achieve the desired moisture content, firmness, and temperature. The characteristics of the produced pellets are presented in Table 1. After cooling, the pellets are moved from the cooler to a holding bunker equipped with a scale via a scraper conveyor. At the holding bunker, pellets are packed into small bags (10, 25, 50 kg), large bags (250, 500, 1000 kg), or containers (20 or 40 feet long). Once packed, pellets are stored at the pellet production facility at the storage warehouses.

Table 1. Characteristics of pellets

Characteristic	Unit	Amount
Moisture content	%	7
Ash content	%	0.5
Low heating value	MJ/kg	17
Bulk density	kg/m ³	630
Carbon content	%	49.70

Source: data obtained from the executive director of the "Russian Pellet Union" association via questionnaire.

It is important to note that the pellet producer can sell pellets directly to the consumer, as it is a part of the company’s business model. The pellets are held in storage containers at the production facility and, when purchased, can be delivered directly to the consumer. Avoiding the additional step of distributing packed pellets from the producer to a dealer provides the pellet producer with greater control over the quality of service, pricing, and customer relationships. This pellet distribution method for pellets is considered and analyzed in our study. The final step is the production of thermal energy at the household level. We assumed the combustion of wood pellets took place in a pellet furnace with a nominal power output of 25 kW, followed by the disposal of ash at a landfill. A 25-kW pellet furnace was selected for this study due to its widespread availability on the Russian market, cost-effectiveness, and sufficient heating capacity for typical residential homes ranging from 200 to 300 m². Pellet combustion efficiency is considered to be 90%. Waste disposal at a landfill was selected as the end-of-life option because the alternative option of utilizing ashes as a fertilizer is not feasible in the region due to the climate conditions limiting private gardening.

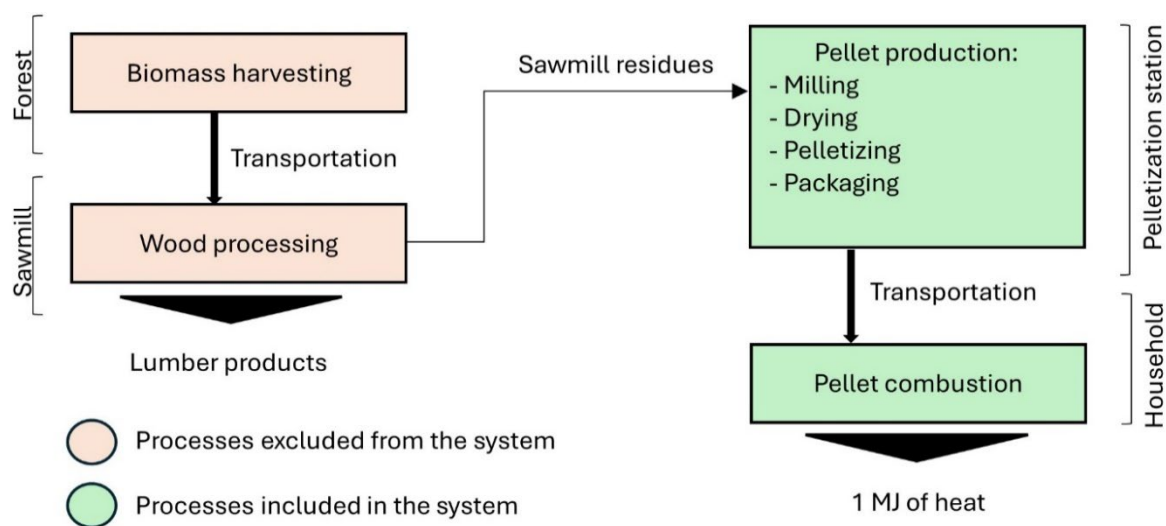


Figure 6. Flow chart of the system boundaries of heat production.

Life cycle inventory

The inventory data used for this LCA are provided by Executive Director of the “Russian Pellet Union” Association and completed with the data from the Ecoinvent 3.4 database (Ecoinvent 2017). Main input and output flows used for the LCA are presented in Table 2. A biogenic carbon neutrality status of biomass used for energy production was applied as it is common practice when conducting LCAs (Ruiz et al. 2018; Quinteiro et al. 2019; Martin – Gamboa et al. 2020). The rationale behind the assumption of carbon neutrality is that the CO₂ emissions occurring during biomass combustion are balanced out by the carbon sequestered during the biomass growth. In this study, the assumption of carbon neutrality is further supported by the fact that sawmill residues not utilized for pellet production and subsequent heat generation would still be disposed of through alternative means, such as combustion or landfilling. In these alternative scenarios, the amount of biogenic emissions released would be comparable to those generated by the biomass being used for pellet production and subsequent combustion. It is crucial to highlight that the concept of carbon neutrality applies exclusively to biogenic emissions and does not encompass non-biogenic emissions, such as those resulting from fossil fuel consumption and other emission sources associated with pellet production and bioenergy

generation. These non-biogenic emissions have been carefully accounted for in our analysis. This concept can also be applied to the drying process and pellets combustion process.

Data for the production of 1 kg of pellets was collected from a primary source via a questionnaire and supplemented with Ecoinvent database (Ecoinvent 2017). The main biomass materials used for pellet production are equal parts sawdust and wood chips, with a small share of mostly softwood shavings. High voltage Russian electricity production mix was used for power generation during the pellet production process. Heat energy, necessary for the drying phase, was generated from the combustion of wood chips in a 50-kW furnace. Wood chips were sourced from wood processing activities at the mill. The Ecoinvent database (Ecoinvent 2017) was used to gather input information for LDPE bags, lubricant oil, and maize starch. The distribution step of the LCA includes transportation of pellets over a distance of 20 km from the pellet producer to the consumer by a light commercial vehicle. This particular distance was selected due to the prevalence of centralized district heating in Russia's national heating system. The Russian centralized district heating system is the largest in the world and serves over 70% of the population (Korppoo and Korobova, 2012). Most of the population lives in apartment buildings in cities. 90% of the heating in urban areas is from centralized heating plants that utilize conventional fossil fuels to produce heat. On the other hand, inhabitants of rural areas live mostly in private houses and only 20% of the heat energy comes from district heating plants. The pellet production facility described in this study is located outside of the general urban area, where district heating ceases to be the default option. Several rural inhabited areas are located around 20 km away from the pellet production facility, hence the decision to select a transportation distance of 20 km. The rationale behind utilizing a light commercial vehicle is its economic feasibility and accessibility. A small number of inhabitants live in rural areas that are relatively close to the pellet mill, which results in short transportation distances and lower quantities of pellets that need to be transported. At the conversion step of the LCA, the production of 1 MJ of thermal energy was calculated. During this step, pellets are directly

combusted in the 25-kW pellet furnace. After the combustion process, it is assumed that the combustion waste will be collected by a state-operated waste management service using a large heavy-duty truck as part of a weekly collection route. The waste will then be transported approximately 30 kilometers from the household to a designated landfill.

Table 2. Inventory data for the production of 1 MJ of heat energy from combustion of pellets in a pellet furnace.

Flow	Unit	Amount	Data description
<i>Pellet production (1 kg)</i>			
Electricity, high voltage	kWh	0.096	Ecoinvent database. Electricity production mix, APOS-U, RU
Heat, central or small, non-fossil	MJ	0.11232	Heat production, wood chips from industry, 50 kW furnace
Lubricant oil	kg	8.4e-5	Ecoinvent database. Lubricant oil, APOS-U, GLO
Maize starch	kg	0.005	Ecoinvent database. Maize starch, APOS-U, GLO.
LDPE bags	kg	0.00228	Ecoinvent database. Packaging film, low density polyethylene, APOS, U - GLO
Saw dust	kg	0.45	Primary data
Wood chips	kg	0.45	Primary data
Wood shavings	kg	0.1	Primary data
Water	m ³	3.0e-5	Primary data
<i>Distribution</i>			

Truck			Freight lorry, light (<3.5 t) commercial vehicle
Distance	km	20	Primary data
<i>Conversion and waste disposal</i>			
Pellets	kg	0.05882	Primary data. Moisture content 7%, Lower heating value 17 MJ/kg
Electricity, low voltage	kWh	0.005	Ecoinvent database. Electricity, low voltage, APOS-U, RU
Furnace			Ecoinvent database. Furnace, pellets, 25kW, APOS, U - GLO
Transportation of combustion waste (ash) to landfill	km	30	Freight lorry, heavy (~25 t) commercial vehicle
Ash	kg	2.9e-4	Primary data

The ISO guideline for LCA (ISO 14044, 2006b) recommends avoiding allocation whenever possible. As the energy and resource demands within the system boundaries of this study were associated solely with the production of a single product, allocation was not performed.

Results and discussion

Life cycle impact assessment

Table 3 shows the results of the impact assessment from the generation of 1 MJ of heat energy. Figure 7 represents the contribution of each step of the life cycle assessment to a particular impact category analyzed.

GW is regarded as the most relevant and extensively studied impact category in bioenergy life cycle assessments (Martin-Gamboa et al. 2020; Musule et al. 2021) due to its direct link to climate change. GW quantifies the greenhouse gas emissions associated with bioenergy systems, providing a clear metric to evaluate their environmental performance and contribution to climate change. Evaluation of global warming from the production of 1 MJ of thermal energy showed emissions of 0.015 kg of CO₂ equivalent (eq). Despite the fact that comparing LCAs can be challenging due to variations in system, temporal, and spatial boundaries, functional units, methodological choices, differences in feedstock, conversion technology, etc., some similarities can be found in LCAs with particular parameters. A review conducted by Martin-Gamboa et al. (2020) showed that GW values from 0 to 0.025 g CO₂ eq use waste biomass as feedstock for pellet production, utilize biomass for drying during pelletization process, assume carbon neutrality of biomass for biogenic emissions, and have low transportation distances. The parameters of the system in our LCA, as well as the GW emissions, overlap with the parameters mentioned in the review. The freshwater eutrophication impact category results show the emission of 5.85e-6 kg of P eq contributed by pellet production (42.5 %) and combustion (49.5 %). Marine eutrophication emissions amounted to 1.42e-6 kg of N eq coming predominantly from the pellet production stage (85.7 %). Terrestrial acidification impact amounted to the emission of 8.29e-5 kg of SO₂ eq and is mainly related to the combustion stage of the life cycle (61.4 %), followed by pellet production (29.2%) and transportation (9.4%). Fine particular matter formation impact category analysis shows the emission of 7.35e-5 kg PM_{2.5} eq generally from the combustion stage (74.5%) and pellet production stage (20.5%). Fossil resource scarcity emissions are equal to 0.0045 kg of oil eq, coming from all three stages of the life cycle in significant shares: pellet production – 49.4%, transportation – 16.7%, and pellet combustion and waste disposal – 33.9%. The analysis of ozone formation (human health) and ozone formation (terrestrial ecosystem) showed practically identical results with the main contributor being the stage of pellet combustion and waste disposal (74%).

Overall, the combustion and waste disposal stage of the life cycle is the most impactful step within the current LCA, as it is the main contributor in five out of the eight impact categories (FE, TA, FPMF, OFHH, and OFTE). The remaining impact categories (GW, ME, FRS) are most influenced by the pellet production step, with transportation step being the least impactful and never exceeding 17% of contribution to any impact category. These findings correspond with the existing literature. For example, Quinteiro et al. (2019) assessed the impact of small-scale thermal energy production in Portugal from the combustion of pellets produced from maritime pine logs. Their findings showed that the main contributor to GW and FRS is pellet production, whereas thermal energy generation has the greatest impact on OFTE, OFHH, and TA. Moretti et al. (2014) analyzed the effects of thermal energy production from pellet combustion in Southern Italy and also concluded GW was the most impacted by the pellet production process. In another study, Topić Božič et al. (2024) conducted an LCA to assess the effects of domestic heat production in Slovenia from combustion of wood pellets produced from sawmill residues. The results of their LCA showed that the pellet production process has the greatest impact on GW even when transportation distances for pellet distribution reach 500 km.

Table 3. Results of impact assessment of production of 1 MJ of heat energy.

Impact category	Unit	Amount
Global warming (GW)	kg CO ₂ eq	0.015
Freshwater eutrophication (FE)	kg P eq	5.85e-6
Marine eutrophication (ME)	kg N eq	1.42e-6
Terrestrial acidification (TA)	kg SO ₂ eq	8.29e-5
Fine particulate matter formation (FPMF)	kg PM _{2.5} eq	7.35e-5
Fossil resource scarcity (FRS)	kg oil eq	0.0045
Ozone formation (human health; OFHH)	kg NO _x eq	0.00012
Ozone formation (terrestrial ecosystems; OFTE)	kg NO _x eq	0.00012

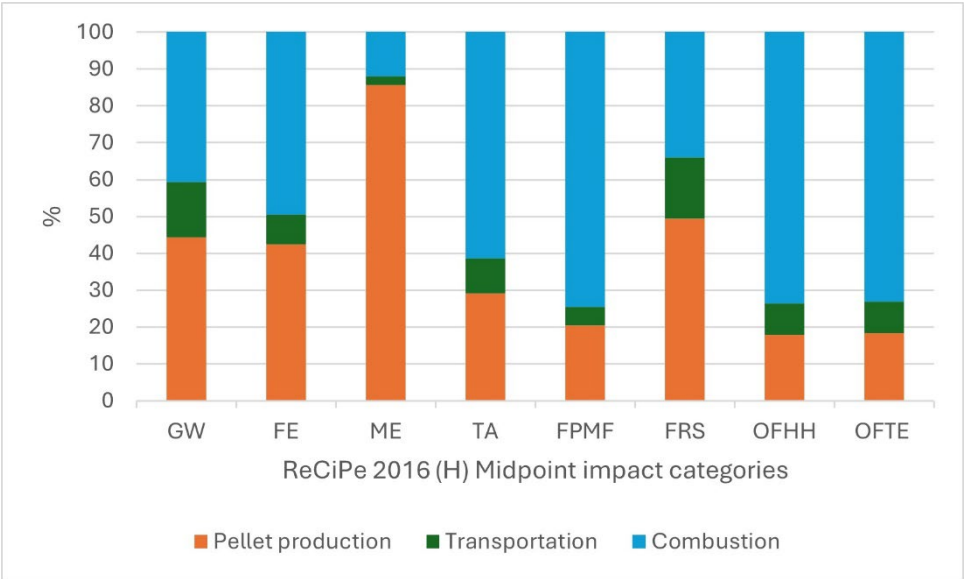


Figure 7. Contribution of each step in LCA by impact category using ReCiPe 2016 (H) midpoint LCIA method. ReCiPe impact categories: global warming (GW), freshwater eutrophication (FE), marine eutrophication (ME), terrestrial acidification (TA), fine particulate matter formation (FPMF), fossil resource scarcity (FRS), ozone formation (human health; OFHH), ozone formation (terrestrial ecosystems; OFTE).

Sensitivity analysis

Three sensitivity analyses were conducted to assess the impact of the methodological assumptions and data used within the framework of this LCA. The first sensitivity analysis consists of the application of an alternative LCIA method to examine whether the same trend in results is observed. The results of the evaluation of ILCD impact categories are presented in Figure 8. Similar trends were observed when analyzing the results of the contribution of each stage of the LCA for all impact categories, with the exception of marine eutrophication. When utilizing the ILCD method, the results showed that the main impact is related to the combustion and waste disposal step of LCA (67%), with pellet production contributing 25%, and the remaining 8% being attributed to the transportation stage. This result is expected considering the difference in methodological approaches these two methods implement in this particular

impact category (Colucci et al. 2021). Such discrepancies in the marine eutrophication impact category come from the higher characterization factors for nitrogen oxides (NO_x), simpler midpoint modelling approach, and lack of regionalization in ILCD 2011 compared to ReCiPe 2016 (European Commission, Joint Research Centre, Institute for Environment and Sustainability 2011; Huijbregts et al. 2017).

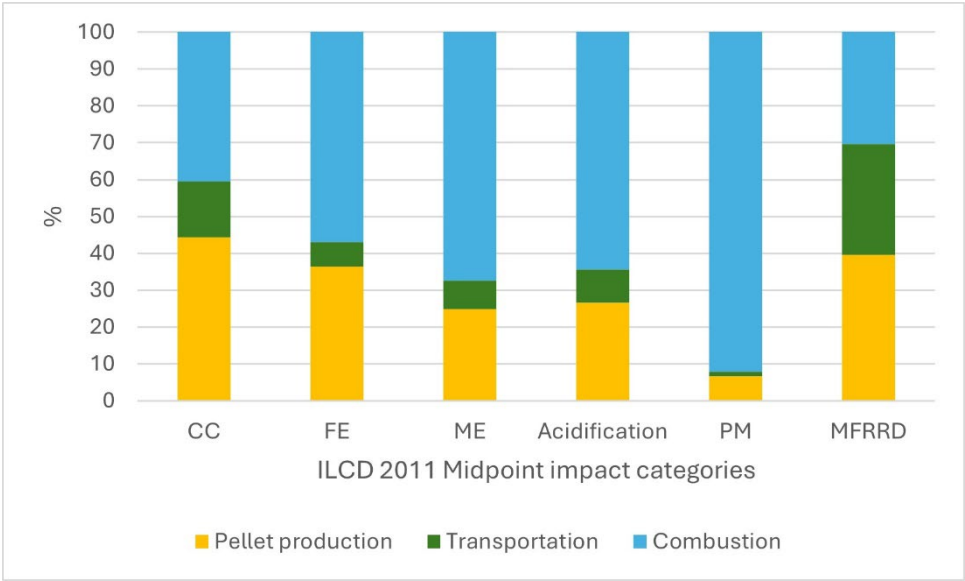


Figure 8. Contribution of each step in LCA by impact category using ILCD 2011 Midpoint method. ILCD Midpoint impact categories: climate change (CC), freshwater eutrophication (FE), marine eutrophication (ME), acidification, particulate matter (PM), and mineral, fossil, and renewable resource depletion (MFRRD).

The second sensitivity analysis consists of evaluation of the alternative source of heat used during the drying stage in the pelletization step of the LCA. Many pellet production facilities in the Northwest region of Russia generate heat for the drying stage of the pelletization by combusting sawmill residues, as is the case in the current study. However, there are facilities that obtain the heat for the drying step from other sources. For this sensitivity analysis, the production of 1 MJ of thermal energy was assessed with the use of natural gas for the drying of biomass for pellet production since it is one of the most common sources of heat in the

Arkhangelsk area in the Northwest region. The results of the impact assessment are presented in Table 4. The GW and FRS impact categories showed a higher number of emissions, compared to the baseline scenario. Both GW and FRS impact categories showed increased emissions by 6.5% and 4.3%, respectively, due to the increased emissions during pellet production. The results of ME and OFTE impact categories remained the same, whereas the overall impact on FE, TA, FPMF, and OFHH decreased. The most significant impact reduction is observed in the OFHH impact category with the amount of NO_x eq emissions decreased by 8.7%. Other impact categories showed lower impact reduction with the results not exceeding 2% reduction. These results correlate with the existing LCA studies that have assessed alternative sources of heat for the drying step of pelletization. The LCA conducted by Padilla-Rivera et al. (2017) demonstrated that the production of one ton of packed pellets using sawmill residues in the form of sawdust, shavings, and woodchips for heating during the drying stage of the pelletization process emits 2.84 kg CO₂ eq, whereas drying using natural gas emits 20.25 kg CO₂ eq. In a study published by Benetto et al. (2015), utilizing wood chips for drying during the pellet production process significantly reduced the impact on GW, FE, and FRS compared to drying using natural gas, grape marc pellets, or electricity from the grid.

Table 4. Sensitivity analysis results by impact category. Baseline scenario – transportation distance of 20km with waste biomass used for drying; sensitivity analysis 2 – natural gas used for drying; sensitivity analysis 3 – transportation distance of 100 km.

Impact categories	Unit	Baseline scenario	Sensitivity analysis 2	Sensitivity analysis 3
		Amount	Amount	Amount
Global warming (GW)	kg CO ₂ eq	0.015	0.016	0.025
Freshwater eutrophication (FE)	kg P eq	5.85e-6	5.82e-6	7.69e-6
Marine eutrophication (ME)	kg N eq	1.42e-6	1.42e-6	1.55e-6

Terrestrial acidification (TA)	kg SO ₂ eq	8.29e-5	8.20e-5	0.00011
Fine particulate matter formation (FPMF)	kg PM _{2.5} eq	7.35e-5	7.22e-5	8.81e-5
Fossil resource scarcity (FRS)	kg oil eq	0.0045	0.0047	0.0075
Ozone formation (human health; OFHH)	kg NO _x eq	0.00012	0.00011	0.00016
Ozone formation (terrestrial ecosystems; OFTE)	kg NO _x eq	0.00012	0.00012	0.00016

The third sensitivity analysis tests the assumed transportation distance. The chosen distance of 20 km in the baseline scenario is due to the proximity of the pellet production facility to the residential areas. For this sensitivity analysis, the distance of 100 km was chosen due to the potential for the pellet production company to increase their distribution networks to smaller rural inhabited areas. Several of these settlements are located further from the center of the Arkhangelsk region at a distance of 100 km from the pellet production facility. Rural communities living further from the urbanized area lack centralized heating and are more likely to have an increased demand for alternative heat sources. The results of this sensitivity analysis can help identify potential environmental impacts and opportunities for optimizing logistics. Additionally, the same light weight truck was kept as the means of transportation as opposed to transportation of pellets by train because it is not an economically viable method for the relatively low demand for pellets in the area associated with the small number of rural inhabitants. The results of LCIA showed significantly increased emissions in all impact categories assessed with the biggest growth of 50% for GW and FRS, and almost 30% for FE, TA, OFHH, and OFTE (Table 4). These results correlate with the existing literature. Topić Božić et al. (2024) observed an increase of 85% in OFHH, 120% in FRS, and 102% in GW if the distance pellets are transported by truck is increased from 100 km to 1000 km. Another LCA

study published by Cleary and Caspersen (2015) showed a 37% decrease in GW related to the transportation of pellets in Canada when the transportation distance is decreased from 1889 km (where 250 km are covered by truck, and 1635 km by train) to 1350 km by train.

The results of this study provide a comprehensive evaluation of the environmental impacts associated with residential heat production from wood pellets in the Northwest region of Russia, utilizing primary data specific to the Arkhangelsk area.

Conclusion

This study evaluated the environmental impact of residential heat production in the Northwest region of Russia using wood pellets derived from sawmill residues through a life cycle assessment (LCA) following ISO 14040 and 14044 standards. The analysis, based on a gate-to-grave approach, included sawmill residue collection, pellet production, pellet distribution, and thermal energy production and ash disposal. The ReCiPe 2016 (H) Midpoint v. 1.1 method was employed to assess the environmental impacts across several categories.

The results indicated that pellet combustion and waste disposal is the most impactful stage in the life cycle, contributing significantly to freshwater eutrophication (FE), terrestrial acidification (TA), fine particulate matter formation (FPMF), and ozone formation (both OFHH and OFTE). Pellet production was the dominant contributor to global warming (GW), marine eutrophication (ME), and fossil resource scarcity (FRS). Transportation had the least impact across all categories.

Sensitivity analyses confirmed the robustness of these findings. When an alternative LCIA method (ILCD 2011 Midpoint) was applied, similar trends were observed, with minor deviations in marine eutrophication impacts due to methodological differences. The evaluation of an alternative heat source for pellet drying (natural gas) resulted in higher emissions for GW and FRS, while other impact categories showed reduced impacts. Increasing the transportation

distance from 20 km to 100 km significantly raised emissions across all impact categories, emphasizing the importance of proximity between the pellet manufacturer or distributor and end users.

This study highlights the environmental benefits and drawbacks of using wood pellets for residential heating in the Northwest region of Russia. While pellet combustion and production stages are the most significant contributors to environmental impacts, optimizing the supply chain, particularly in terms of transportation distance and heat sources for drying, can mitigate some of these impacts. The findings provide valuable insights for policymakers and stakeholders aiming to enhance the sustainability of bioenergy systems.

The importance of aspects of the transportation process, as highlighted in the current study, underscores the need for future LCA studies to investigate alternative transportation methods and transportation distances. Additionally, subsequent studies could compare the findings of this study with those of other bioenergy production scenarios. These scenarios might include:

- The use of different biofuels, such as wooden chips or firewood, to assess their comparative environmental impact.
- The use of various biomass sources, such as logging residues or bioenergy plantations, used for biofuel production to identify the most suitable feedstock options.
- The use of different conversion technologies, such as pyrolysis or gasification, to determine their potential for emission reduction.

The findings from such comparative analyses can provide critical insights for policymakers, enabling them to make more informed decisions on climate change mitigation strategies.

Chapter 5: Methods for monetary valuation of ecosystem services: a scoping review

5.1. Summary and contribution statement

Citation of the article: Selivanov E., Hlaváčková P. (2021): Methods for monetary valuation of ecosystem services: A scoping review. Journal of Forest Science

DOI: <https://doi.org/10.3832/for4160-015>

As a lead author of this scoping review on methods for monetary valuation of ecosystem services, I was responsibilities included: design of the framework of the study; development of the search strategy to identify relevant literature; screening and selecting of the studies; extraction, synthesis, and analysis of the data; and interpretation of the findings. In addition, I wrote the manuscript, created tables and figures, and managed the publication process.

The main objective of this article was to identify methods that can be used for monetary evaluation of ES via a scoping review methodology.

The results of our scoping review present a description of methods for monetary valuation that were classified into three main groups: direct market valuation methods, indirect market valuation methods, and non-market valuation methods. Direct market valuation methods include market-price-based approach, cost-based approach, and product function valuation approach. Indirect market valuation methods consist of travel cost and hedonic pricing approaches. Non-market valuation methods include contingent valuation, choice modelling, and group valuation methods. Additionally, the following methods that are not included in three main groups were identified and described: shadow pricing approach, value (benefit) transfer methods, resource rent method, and simulated exchange method. Our study identified several gaps in the communication of valuation methods, including inconsistencies in terminology and hybridization (conflation of two distinct concepts under a single method).

5.2. Content of the article “Methods for monetary valuation of ecosystem services: a scoping review”

Introduction

The concept that people benefit from the environment has been accepted for a very long time. During the development of the field of environmental science, this concept has become known as ecosystem services (ES). The term was used for the first time in an article published by Ehrlich and Ehrlich (1981). In 1997, two major works were published (Costanza et al. 1997; Daily 1997) that served as catalysts to new research and policies related to this topic. In 2005, the Millennium Ecosystem Assessment (MEA) proposed a classification of ES and brought more attention from policy makers (MEA 2005). Even though the classification of ES was adjusted and developed in later publications (TEEB 2010; Haines-Young and Potschin-Young, 2018), the original classification is widely used by researchers even today. According to MA, ES can be divided into four categories. The first category is provisioning services, which includes food, timber, fresh water, and other products obtained from ecosystems. Next, regulating services are benefits obtained from the regulation of ecosystem processes such as climate regulation and water filtration. The third category is cultural services, or nonmaterial benefits derived from ecosystems including recreation, tourism, aesthetics, and spirituality. The last category is supporting services that represent services necessary for the production of all other ecosystems – soil formation, nutrient cycling, etc. In addition to the categorized definition, it is important to understand the difference between ES and ecosystem functions. Ecosystem functions can be defined as the capacity of natural processes and components to provide goods and services that satisfy human needs, directly or indirectly (DeGroot et al. 2002). However, ecosystem functions exist independently of human needs. In contrast, ES require humans to appreciate the goods and services provided by ecosystems. Despite the fact that there is no single accepted definition, ES can be defined as direct and indirect contributions of ecosystems

to human well-being (TEEB 2010). Contributions of ecosystems can also be referred to as benefits or values. According to the Total Economic Value (TEV) framework (Figure 9), benefits that humans obtain from ecosystems can be divided into use values and non-use values. Even further, use values consist of direct use values, indirect use values, and option value. Direct use values are goods and services that can be consumed directly (timber, medicinal plants, etc.). Indirect use values are goods and services that are enjoyed indirectly but they can contribute to another activity (crop pollination, carbon sequestration, etc.). Option value is the benefit placed on the potential ability to use a resource in the future even though it is not currently being used, while the likelihood of future use is very low (Conner 2014). Non-use values include bequest value and existence value. Bequest value can be defined as the value attributed to maintaining something for the benefit of future generations. Existence value is the satisfaction from knowing that something exists.

Adequate assessment of ES is important for the development of policies and management plans related to forestry activities and the environment. Carefully identified ecosystem values can determine which options policy makers should prioritize to provide the greatest benefit. It is particularly relevant when planning forest management activities because forests, when sustainably managed, simultaneously fulfil ecological, economic, and social functions, and provide a great number of supporting, regulating, cultural, and provisioning ES that significantly affect human well-being (Fürst et al. 2007). Over recent years, the influence of the ES framework on environmental and conservation policy has grown (Kull et al. 2015). Generally, there are three different ways to assess the value of ES: qualitative analysis, quantitative analysis, and monetary analysis (TEEB 2011). According to Kettunen et al. (2012), qualitative analysis focuses on non-numerical indicators of the value such as benefits to mental and physical health, social benefits from recreation. Quantitative analysis focuses on numerical data such as quantity of sequestered carbon, quality of water, etc. Monetary analysis focuses on translating the qualitative and quantitative aspects into a particular currency. Monetary

valuation of ES is the most widely applied approach (Christie et al. 2012), as it is often deemed to be the most pragmatic language when it comes to communication with political and business institutions (Spash 2013).

Despite the fact that there is neither commonly accepted methodology nor statistical standards for ES assessment (Whitham et al. 2015; Kornatowska, Sienkiewicz 2018; Mengist, Soromessa 2019), the number of published articles dedicated to ecosystem services and, in particular, to the assessment of ES is increasing (McDonough et al. 2017; Acharya et al. 2019). Furthermore, the number of studies devoted to the assessment of forest ES has gradually increased in recent years (Mengist, Soromessa 2019; Di Franco et al. 2021). There are numerous methods used by researchers to evaluate ES. Especially, monetary valuation can be carried out by means of a wide variety of different approaches. The main goal of this review is to analyse available literature using the methodology particular to the scoping review approach. This approach facilitates the identification of all possible methods that can be used for monetary valuation of ES. It should be noted that in this paper, the terms ‘method’ and ‘approach’ will be used interchangeably. Similarly, the terms ‘valuation’ and ‘assessment’ will be treated as synonyms.

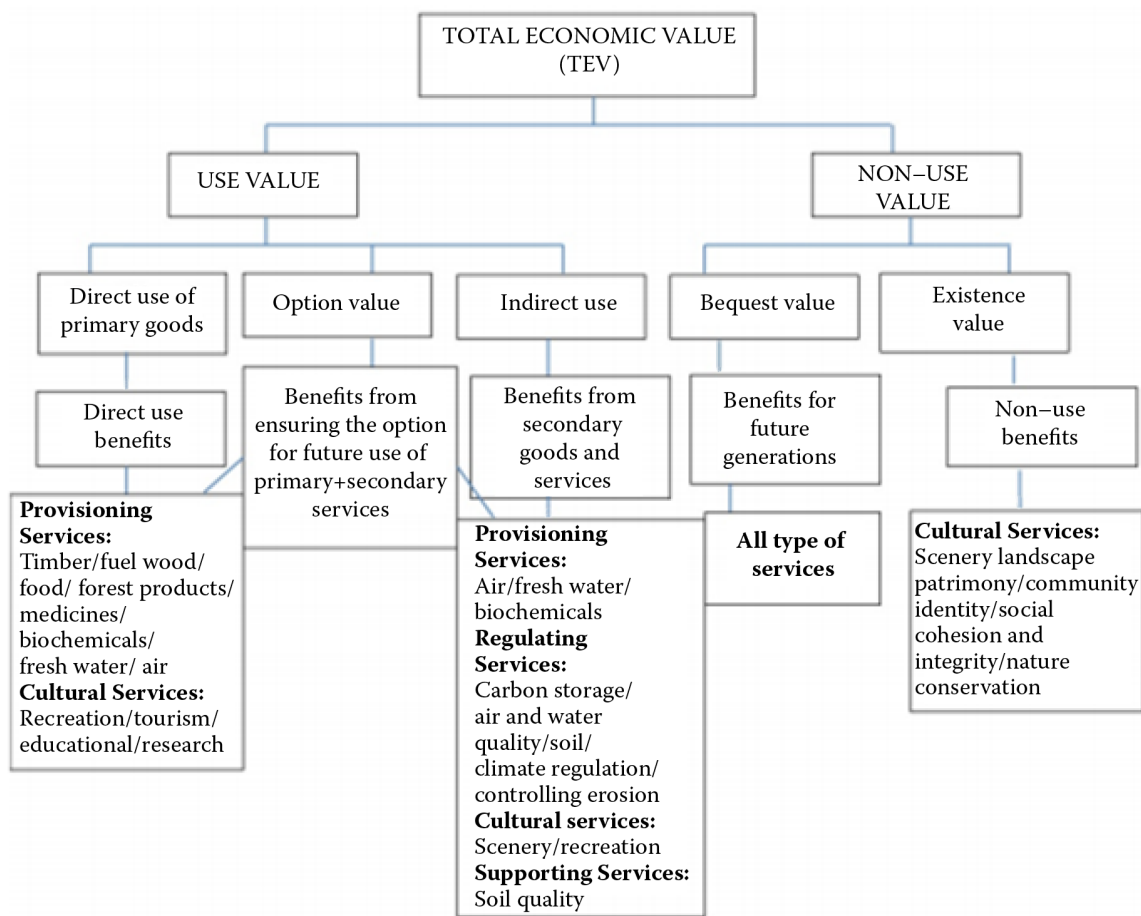


Figure 9. Total economic value framework (Mendes 2012)

Methods

Scoping review

The review presented here was conducted according to the scoping review methodology. This methodology is appropriate for identifying and analyzing knowledge gaps, for clarifying general concepts and definitions in literature, and for identifying key topics within a field of research (Munn et al. 2018). The approach facilitates the analysis of existing literature and a summary of findings from a range of different study designs and methods. This paper will use the five original stages of the scoping review methodology first introduced by Arksey and O'Malley (2005), but it will also follow recommendations derived from Guidance for the Conduct of JBI Scoping Reviews (Peters et al. 2020). The five stages are: (1) identifying the

research questions, (2) identifying relevant studies, (3) selecting eligible studies, (4) charting the data, (5) and summarizing and reporting findings.

Identifying research questions.

The main research question of this scoping review is: what are the methods that can be used for monetary valuation of ecosystem services?

Identifying relevant studies. The Web of Science and Scopus databases were used in order to identify studies relevant to the purpose of this scoping review. The publication time frame entered into the search criteria for each database ranged from 2010 to March of 2021, with the time frame ending at the time this review was performed. 2010 was chosen as a starting point for the literature review because the quantity of articles focusing on monetary assessment of forest ecosystem services rapidly increased starting that year compared to the consistently lower number in previous years (Acharya et al. 2019). The search strategy involved a data search by title in both databases using the following key phrases: “ecosystem services assessment method” and “ecosystem services valuation method”. In addition, a forward-backward article search (also known as ‘citation chaining’) was used to collect references that are frequently cited in topical papers (Robinne et al. 2020). Furthermore, a list of inclusion criteria was developed for the study selection stage:

- Peer-reviewed articles, published book chapters, and reports were analyzed.
- All analyzed publications were written in English.
- No restrictions regarding the country of publication were enforced.
- Reviewed articles were not limited by the ecosystem type.
- All analyzed articles needed to include a description of method(s) used for monetary valuation of ecosystem services.
- Case studies that had a thorough description of the valuation method were also included in the scoping review.

Study selection.

In the study selection process, a total of 109 articles were identified using the chosen key phrases. Out of these 109 articles, 59 were found via Web of Science (WoS) database and 45 articles were found via Scopus database. During the screening process, 49 papers were excluded as not relevant to the topic and 18 papers were excluded as duplicates. Therefore, 42 articles were obtained from WoS and Scopus; five additional articles were obtained through chaining. As a result of the screening, 42 articles were selected for subsequent full-text analysis. In the course of the full-text analysis, 26 articles were excluded. Therefore, the final number of articles included in this scoping review is 16. The study selection process is summarized in Figure 10.

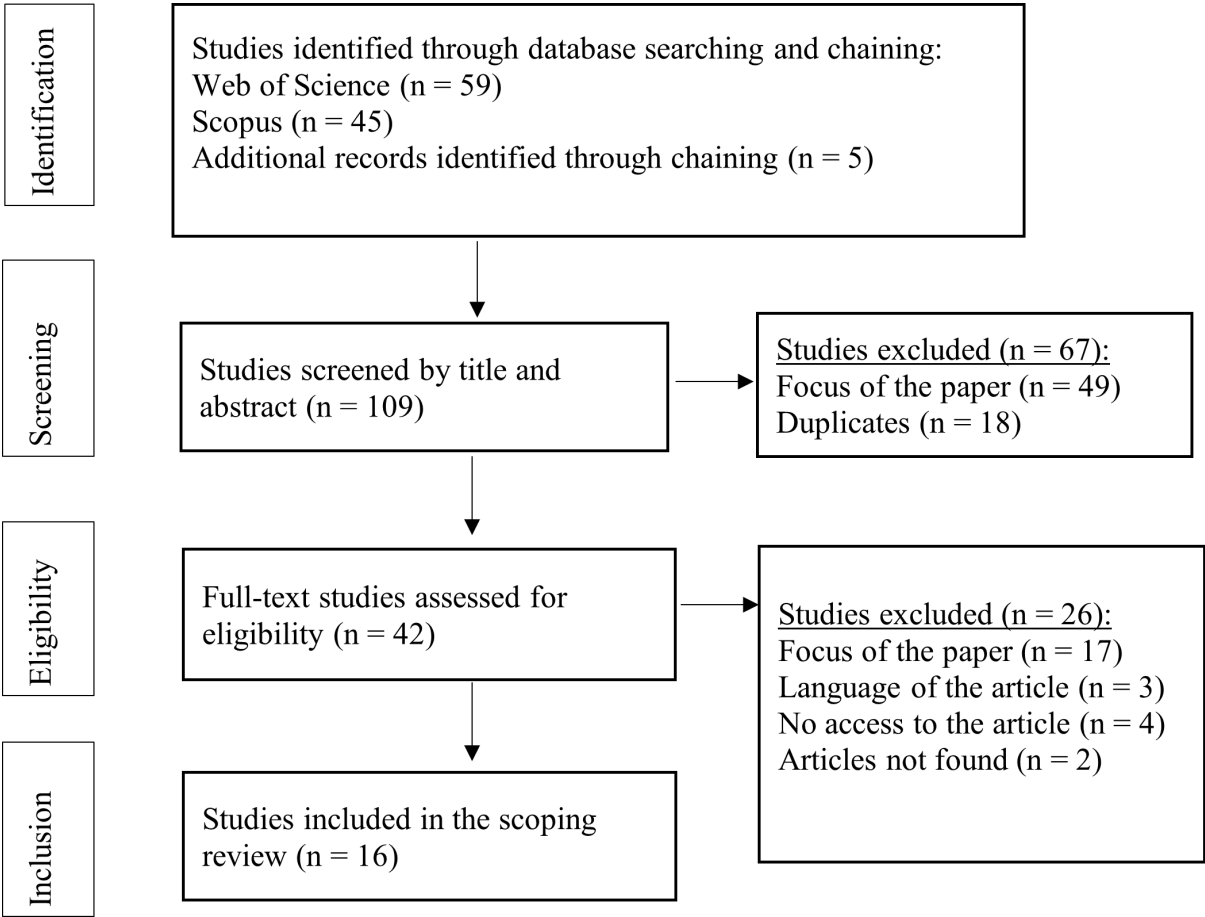


Figure 10. Flow chart of the study selection process

Charting the data.

Charting the data provides a logical and descriptive summary of the results that aligns with the objective and question of the scoping review (Peters et al. 2020). During this stage of the scoping review, the charting table was developed in order to record the key information that was relevant to the objectives of the study. This table included the following information about analysed articles: author, year of publication, type of publication, goal of the study, definition of the 'value', name of the method for ecosystem service valuation, description of the method, and other notes.

Summarizing and reporting findings.

The last stage of the scoping review is summarizing the data in relation to the purpose of the review, making conclusions, and noting any implications of the findings.

Results and discussion

Study context

A total of 16 articles were included in this scoping review. Nine of them were published within the last five years. Most of these papers (75%) were published as peer-reviewed articles, whereas two papers were published as book chapters and one paper as a report. Regarding the goal of these studies, eight publications mentioned a review of the ES valuation methods among the objectives of the paper (e.g., Arias-Arévalo et al. 2018; Van Beukering et al. 2015). Two articles were focused on reviewing only one particular valuation method (Chang, Yoshino 2017; Navrud, Strand 2018). Other papers did not state reviewing valuation methods among their main goals. Nevertheless, these articles provided an overview of at least one ES valuation method.

Definition of the term ‘value’

During the process of data charting, one of the investigated issues was the definition of the term ‘value’ as provided by the authors. Analysis of the literature showed that ten articles did not provide a definition of the term ‘value’. Other articles defined the term ‘value’ similarly as ecosystem service contributions to human well-being (Costanza et al. 2017; Chen 2020). In addition, some articles discussed the concept of the value in more detail and provided multiple definitions of ‘value’ (Hattam et al. 2015; Costanza et al. 2017; Kornatowska, Sienkiewicz, 2018).

Monetary valuation methods identified

During the full-text analysis of the articles included in this scoping review; many methods applicable to monetary valuation of ecosystem services were identified. Table 5 summarizes all valuation methods identified during the scoping review.

Description of ES assessment methods.

The following part of this paper aims to describe the identified ES assessment methods. In this study, valuation methods are divided into three main groups according to Van Beukering et al. (2015): direct market valuation methods, indirect market valuation methods, and non-market valuation methods (Figure 11).

Direct market valuation methods

This group of methods uses data from existing markets as a basis for the ES valuation process. That means that this method operates with prices for goods and services that actually exist in real markets. Direct market valuation methods consist of the following approaches: market price-based (exchange-based) valuation approach, cost-based valuation approach, and production function valuation approach.

Market price-based approach (aka direct market prices valuation approach or exchange-based valuation). This approach uses the prices of ES that are traded on the markets as a proxy for their monetary value (Arias-Arévalo et al. 2018). For example, the value of the timber

production ES can be defined as the price of this timber on the market. When market prices are adjusted to correct for market distortions such as taxes and subsidies, this approach could be called adjusted market price method (Bateman et al. 2011).

The cost-based valuation approach assumes that the value of ES can be defined at least as the costs that are avoided because ES exist. There are a few variations of the cost-based approach: replacement cost method, substitute cost method, damage cost avoided method, and mitigation cost method.

- Replacement cost method estimates the value of an ES as the costs associated with replacing it. For example, if the forest stand has to be harvested due to construction work, the value of the forest stand will be equal to at least the costs of establishing a new forest stand by planting new trees.

- Substitute cost method is recognized as an original method that differs from replacement cost approach (Van Beukering et al., 2015; Kornatowska, Sienkiewicz 2018). However, most of the articles included in the scoping review do not distinguish between these two methods and both replacement cost and substitute cost concepts are summarized under the umbrella of replacement cost method. Nevertheless, substitute cost approach can be defined as a method that uses the cost of the ES substitute as the value of this ecosystem. For example, the costs of building a water storage tank as a substitute to a lake.

- Damage cost avoided method (aka preventive expenditure method or averting cost method) defines the value of an ES as the costs associated with the hypothetical damage that was avoided due to the existence of this ES. A classic example is a mangrove forest that prevents the damage that could have been caused by storm waters.

- Mitigation cost method (aka restoration cost method) assumes that the cost of ES is equal to the expenses incurred from the mitigation of the negative effects caused by the degradation of the ecosystem. An example would be the purification of the water from a polluted reservoir that

provides drinking water. The main difference between this method and damage cost avoided method is that the damage is hypothetical in the latter.

Production function approach (aka net factor income approach) assumes that the value of an ES can be estimated as the contribution of this ES to the enhancement of income or productivity (Pascual et al. 2010). For example, the value of an ES such as water purification can be estimated based on the increased revenues from selling better quality water on the market.

Table 5. Methods for monetary valuation of ecosystem services identified in the analyzed articles

Publication	Methods described
Arias-Arevalo et al. (2018)	Market price based: market prices Market cost based: replacement costs; damage cost avoided; production function Revealed preference methods: travel cost; hedonic price Stated preference: contingent valuation; choice modeling Benefit transfer Deliberative economic valuation
Chang and Yoshino (2017)	Contingent valuation: WTP; WTA; WTS
Chen (2020)	Revealed preference: travel cost; market method Stated preference: contingent valuation; choice experiment Cost based methods: replacement cost; avoided cost Group deliberation or system modeling The benefit transfer method
Chu et al. (2020)	Revealed preferences: travel cost; hedonic pricing Stated preferences: choice experiments; contingent valuation (WTP; WTA)
Harrison et al. (2018)	Cost-effectiveness analysis: benefit cost Market price/exchange-based method: shadow pricing; mitigation cost-based valuation (restoration cost; replacement cost; clean-up cost)

	Revealed preferences: travel cost; hedonic pricing Stated preferences: contingent valuation; choice experiment; contingent ranking Resource rent method Simulated exchange Production/cost function Value transfer/benefit transfer
Hattam et al. (2015)	WTP
Kornatowska and Sienkiewicz (2018)	Stated preferences: contingent valuation (WTP; WTA); choice experiment Revealed preferences: travel cost; hedonic pricing; benefit transfer; cost-based methods (damage cost avoided; replacement cost; substitute cost)
Van Beukering et al. (2015)	Market based valuation: direct market prices; production function; Replacement costs; substitute costs; avoided damage costs Revealed preferences (Indirect market valuation): hedonic pricing; travel cost Stated preferences: contingent valuation (WTP; WTA); choice experiments Meta-analysis Value transfer
Navrud and Strand (2018)	The Delphi method
Torres-Miralles et al. (2017)	Contingent valuation: WTP Inferred valuation method
Villalobos and Huenchuelo (2010)	Contingent valuation: WTP Choice experiments
Costanza et al. (2017)	Revealed preferences: replacement cost; hedonic pricing Stated preferences: contingent valuation; choice experiments (conjoint analysis)
Costanza et al. (2011)	Revealed preferences: production-oriented analysis (production function); travel cost; hedonic pricing Stated preferences: contingent valuation; choice experiments

Bateman et al. (2011)	Adjusted market prices Production function Damage cost avoided Averting behavior Revealed preferences: travel cost; hedonic property price Stated preference: WTP; choice experiments
Forest Europe (2014)	Preference based valuation: travel cost; contingent valuation; choice experiments; hedonic pricing; market observations Cost-based methods: preventive expenditure (averting costs); replacement costs; damage costs
Pascual et al. (2010)	Direct market valuation methods: market price-based approaches; cost-based approaches (avoided cost; replacement cost; mitigation/restoration cost); production function Revealed preference approaches: travel cost; hedonic pricing Stated preference approaches: contingent valuation (WTP; WTA); choice modeling; group valuation (deliberative monetary valuation) Benefit/value transfer (BT): unit BT; adjusted unit BT; value function transfer; meta-analytic function transfer

WTP – willingness to pay; WTA – willingness to accept; WTS – willingness to sell

Indirect market valuation methods

Indirect market valuation methods or the revealed preference approach assumes that ES values can be “revealed” through observable consumer behaviors or activities in relevant markets (Chu et al. 2020). This approach, in contrast to the stated preference approach described below in the text in this paper, uses data obtained from real markets where actual transactions have been made. During the analysis of the literature included in this scoping review, two indirect market valuation methods were identified: travel cost method and hedonic pricing method.

The travel cost method is commonly used for assessment of recreational ES. The rationale behind this method is that the base value of a particular ES can be measured as the time spent and costs incurred to access a recreational service. For example, the value of hiking at the national park can be reflected in the costs and time spent to travel to the park.

The hedonic pricing method assumes that the value of ES can be at least partially captured in the price of a property. The main idea is that there is a direct relationship between the location of real estate and its price. The closer the house to an ES, the higher the price of the house. For example, the difference between the price of a house located near a lake and the house located in the same area but farther from the lake can be interpreted as the value of ES provided by the lake.

Non-market valuation methods

The non-market valuation approach is often referred to as the stated preference approach. This approach can be used when no market prices are available, and it is not possible to apply methods particular to the revealed preference approach (Van Beukering et al. 2015). The stated preference approach uses the hypothetical choices individual respondents stated to estimate the change in utility associated with a proposed increase in quality or quantity of an ES or bundle of services (Bateman et al. 2002). Non-market valuation methods include contingent valuation methods, choice modelling, and group valuation.

Contingent valuation (CV) methods have four variations:

- Willingness to pay (WTP) – this method assesses the value of an ES by directly asking people how much they would be willing to pay for ecosystem improvements and the ES they will generate (Costanza et al. 2011; Endalew et al. 2020).
- Willingness to accept (WTA) – the WTA method asks individuals how much money they are willing to accept as a compensation for the loss or degradation of the ecosystem and goods and services it provides.

– Willingness to sell (WTS) – this method was described in detail by Chang and Yoshino (2017). Just like WTP and WTA, this method directly questions respondents to define what the reasonable price of some ES may be. This method tries to evaluate ES by asking respondents how much the good or service provided by an ecosystem is worth if sold or how much a third party would be willing to pay for this good or service. Unlike WTP and WTA, WTS approach attempts to keep the respondent in a neutral position. If the respondent has a closer relationship with the side of the seller during the survey, the answer will likely lead to an overestimated value. If the respondent relates more closely to the purchaser, the answer would likely be more cost-conservative. Thus, this method tries to take into account the perspective of both the seller and the buyer equally. In the application of this method, the question asked of participants can be formulated as follows: What would be the reasonable price for ES for both the buyer and the seller? Chang and Yoshino (2017) provided examples of the questions for assessment of ES by using three different methods:

(1) WTP question: If the construction of a new park like Amoy Bay Park requires public donations, what is the maximum amount of money you are willing to donate for the new park?

(2) WTS question: If this Amoy Bay Park belonged to one of your friends, and for some reason he wanted to sell the park to another one of your friends, how much money do you think the park was worth if you were the middleman? What price do you think may be reasonable for both the buyer and the seller?

(3) WTA question: If this Amoy Bay Park belongs to you, it is going to be demolished for other land use purposes, and you will not be able to use it in the future, what is the minimum financial compensation you would expect for the loss of the park?

– Inferred valuation method – this method was mentioned once by Torres-Miralles et al. (2017) during the literature analysis. The inferred valuation method is similar to WTS as it attempts to minimize people's biases when answering standard questions particular to WTP and WTA. Authors of this method (Lusk, Norwood 2009) believe that in order to avoid biases, it would be

the most effective to ask people to predict or infer others’ values for an ecosystem service or good it provides instead of asking people to state their own opinion of its value.

Choice modelling (CM) is a group of survey-based methodologies for modelling respondent’s preferences for goods and services, which are described in terms of their level of certain attributes (Hanley et al. 2001). CM method, similar to CV method, operates with hypothetical scenarios. According to this approach, the value of ES can be assessed based on the respondent’s WTP stated during the process of ES ranking, rating, or choice from a list of alternatives. Choice modelling methods identified during the scoping review in this paper include choice experiment and contingent ranking.

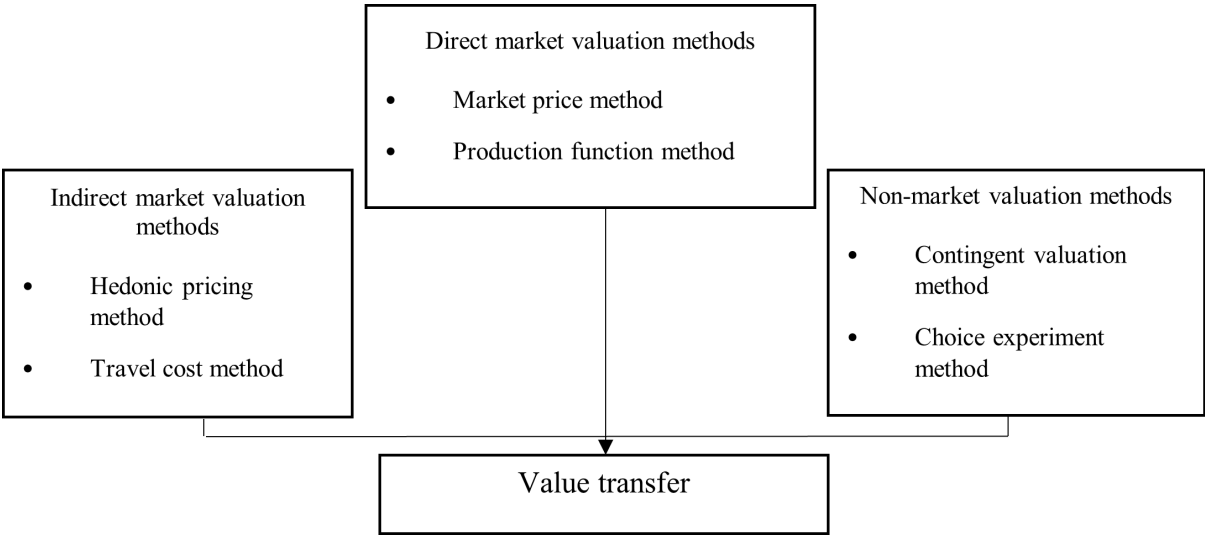


Figure 11. Methods used for valuation of ecosystem services (Van Beukering et al. 2015)

- Choice experiment method (conjoint analysis) estimates the value of ES through a questionnaire in which respondents have to choose one option from a list of alternatives.
- Contingent ranking method asks respondents to rank a set of alternatives given in the questionnaire.

Group valuation (deliberative valuation) methods assess ES based on the stated preferences obtained during the deliberation process of a group of individuals, as opposed to aggregated preferences of individuals gathered during CV and CM approaches. The main idea of the group valuation method is to bring small groups of stakeholders together to discuss and debate the

relative importance of a particular set of public goods (Murphy et al. 2017). This method allows participants to evaluate ES based on their knowledge and understanding of the trade-offs identified during the discussion.

The Delphi method is a variation of the group valuation approach. This method was mentioned only in one paper (Navrud, Strand 2018) during the process of the literature analysis. The basis of this approach is similar to the traditional group valuation method, with the addition that the deliberation process for the value of ES occurs in the group of experts. Navrud and Strand describe the process of Delphi method as follows:

A common approach is to face experts with two or more rounds of questionnaires. After each round, a facilitator provides an anonymous summary of the experts' forecasts from the previous round as well as the stated reasons for their judgments. Thus, experts are encouraged to revise their earlier answers in light of the replies of other members of their panel.

Similar to the group valuation methods, the Delphi method is used to describe the preferences regarding ES stated by a group after the process of discussion amongst the individuals, rather than the preferences stated by individuals as a result of CV and CM.

Other methods

In addition to the methods described previously in this paper, there are methods that are difficult to assign to a particular group of methods. However, these uncategorized methods were also mentioned in the analyzed articles as options for evaluating ES. These methods include shadow pricing approach, value or benefit transfer methods, resource rent method, and simulated exchange method.

The shadow pricing method assesses the value of an ES based on the implicit form of market price, which is defined as the marginal price that society puts on the provision of non-marketed ecosystem services by setting environmental targets (Konrad et al. 2017). For example, when a country signs the Water Framework Directive, the costs that will be incurred in order to reach

the agreed level of water quality can be used as data for the implicit value of water quality improvements (Kelemen et al. 2015).

The benefit transfer method (value transfer method) is not technically a valuation method (Van Beukering et al. 2015). However, it is an approach that can be used to estimate the value of ES. The main idea is that ecosystem values obtained at previously estimated sites (study sites) can be adjusted and extrapolated to a new site (policy site) that possesses similar socioeconomic and ecological contexts. Pascual et al. (2010) distinguished four variations of benefit transfer (BT): unit BT, adjusted unit BT, value function BT, and meta-analytic function transfer. They are characterized as follows:

- Unit BT method estimates the value of an ES at a policy site by multiplying a calculated mean unit value obtained from a study site by the quantity of that ecosystem service present at a policy site.
- Adjusted unit BT approach involves making corrections to the transferred unit values in order to reflect the individual differences between study site and policy site.
- Value function transfer uses functions estimated via valuation applications (travel cost, hedonic pricing, CV) for a study site together with information on parameter values for the policy site in order to transfer values. Parameter values of the policy site are plugged into the value function to calculate a transferred value that better reflects the unique characteristics of the policy site.
- Meta-analytic function transfer (meta-analysis) utilizes a value function estimated from multiple study sites pooled together with information on parameter values for the policy site to estimate values. Therefore, the value function is not derived from a single study but from a collection of studies.

Resource rent method – this method was mentioned only once in the analyzed literature (Harrison et al. 2018). The rationale behind this method is that the value of ES can be derived as a residual amount after the contributions of other forms of capital have been deducted from the

operating surplus. Since the residual reflects the return to the ecosystem asset that is used in production of marketed goods, it is consistent with exchange values (Obst et al. 2015).

Simulated exchange value method – this method was also described only once (Harrison et al. 2018) during this scoping review. The simulated exchange method consists of using WTP data that are estimated using the non-market valuation methods to simulate the entire market (demand, supply and competitive environment) to obtain the market value that one could obtain from a given ES if it were internalized into the market (Caparrós et al. 2017).

All ES valuation methods identified in this scoping review are presented in Figure 12.

Direct market valuation methods	Indirect market valuation methods	Non-market valuation methods	Other methods
• Market price-based approach (e.g. adjusted market prices) • Cost-based valuation methods: • replacement cost • substitute cost • damage cost avoided • mitigation cost • Production function	• Travel cost method • Hedonic pricing method	• Contingent valuation methods: • willingness to pay • willingness to accept • willingness to sell • inferred valuation method • Choice modelling methods: • including choice experiment • contingent ranking method • Group deliberation (e.g. Delphi method)	• Shadow pricing • Benefit transfer method: • unit BT • adjusted BT • value function transfer • meta- analysis • Resource rent method • Simulated exchange method

Figure 12. Methods for monetary valuation of ES identified during the scoping review

Conclusion

This scoping review provides a number of methods that can be applied to the monetary valuation of ES. Over twenty valuation approaches were identified throughout the scoping review. Even though the majority of the methods are widely known (e.g., WTP, WTA, travel cost method, replacement cost method, etc.), several less common methods (e.g., WTS, Delphi method) were identified during the literature analysis. In the process of compiling the range of

Chapter 6. Is forest biomass used for bioenergy production a viable climate change mitigation option?

6.1. Summary and contribution statement

Citation of the article: Selivanov E., Krásná K., Cudlín P. (2025). Je využití lesní biomasy pro produkci bioenergie účinným nástrojem pro mitigaci klimatu? (Is forest biomass used for bioenergy production a viable climate change mitigation option?) Zprávy lesnického výzkumu 70 (4): 231-236.

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As the lead author of this short communication paper, my contribution included establishing the framework of the overall structure of the article, comprehensive synthesis of the data, and communication of the findings. Additionally, I was responsible for writing the complete manuscript.

This publication is structured as a short communication, synthesizing findings from a scoping review of the carbon neutrality concept and an LCA of heat production articles, culminating in an overarching conclusion. The main goal of this article was to concisely present the key insights from articles presented in Chapters 3 and 4 in an accessible format as well as to discuss the CN status of forest biomass used for bioenergy through an LCA perspective.

This short communication presents the main results from two complementary studies: a scoping review and an LCA study. The synthesis of these two articles reaffirms the idea that forest biomass can serve as an alternative energy source for CC mitigation strategy if used sustainably, although its CN status is not inherent but rather context dependent. The findings suggest that the CN of forest biomass can be affected by various factors, such as temporal and

spatial boundaries, and bioenergy production practices (e.g. feedstock type, conversion technology, transportation, etc.). By integrating insights from both studies, this paper highlights the need for standardized methodological frameworks to ensure consistent evaluations of biomass energy systems. Additionally, it proposes directions for future research.

6.2. Content of the article “Is forest biomass used for bioenergy production carbon a viable climate change mitigation option?”

Carbon neutrality of forest biomass

Recently, the perception of carbon neutrality has been questioned within the scientific community (NORTON et al. 2019). Carbon neutrality is a very broad and vague term and currently there is no widely accepted definition. The general idea of carbon neutrality, in the context of bioenergy, is that biomass harvested and combusted for energy production does not add any additional carbon to the atmosphere. A scoping review (SELIVANOV et al. 2023), investigated the problem of carbon neutrality of forest biomass derived from harvesting and sawmill residues, bioenergy plantations, clearcuts, etc., in order to distinguish the most widely used definitions of carbon neutrality and identify the most frequently discussed aspects of the carbon neutrality concept. During the scoping review that included the analysis of 39 articles, eight different concepts of carbon neutrality were identified. These concepts are presented in the Table below.

Table 6. Carbon neutrality concepts

Name of carbon neutrality concept	Core idea	Explanation
Carbon cycle neutrality	Harvested forest biomass that is subsequently replanted achieves carbon neutrality.	Carbon simply circulates between atmosphere and biosphere. When biomass is burned, emissions released to the atmosphere create “carbon debt” which can be “paid back” when the same amount

		of carbon is reabsorbed during the regrowth.
Inherent carbon neutrality	Biomass combustion returns carbon to the atmosphere that was recently absorbed during the process of plant growth.	During the biomass growth “carbon credit” is established and when biomass is combusted, carbon is released back to the atmosphere which creates a balance between absorbed and emitted carbon.
Offset carbon neutrality	Emissions related to harvesting in a forest stand (plot-level) can be offset by carbon sequestration in other stands within the same forest (landscape-level).	If the annual amount of carbon released from harvesting activities in the forest is less than or equal to the annual amount of carbon sequestered during the process of forest growth, then there is no net reduction in carbon stock.
Carbon neutrality of forest residues and waste	Parity of emissions in both bioenergy production and natural decomposition scenarios.	Carbon released during the process of combustion is equal to the amount of carbon that would otherwise be released into the atmosphere via the process of natural decomposition
Carbon neutrality of additional biomass	Bioenergy plantations create carbon credit.	Biomass that has been planted with the sole purpose of being used for bioenergy production is carbon neutral because if not planted, carbon that was sequestered during plant growth would otherwise remain in the atmosphere.
Accounting carbon neutrality	Carbon emitted during biomass combustion was previously accounted for when biomass was harvested.	Based on Kyoto Protocol and UNFCCC principles (UNFCCC 2020), emissions from biomass combustion should be counted as zero because these emissions must be reported in the Land Use, Land Use Change and Forestry category, thus allowing to avoid double-counting.
Substitution carbon neutrality	Net emissions from bioenergy production are smaller than those of fossil fuels.	Net emissions released during the process of biomass combustion are less than emissions associated with the exploitation of alternative energy sources, such as fossil fuels
Life cycle carbon neutrality	Concept considers emissions over the life cycle.	Biomass can be called carbon neutral if it does not release net carbon emissions into the atmosphere when all stages of the life cycle of bioenergy production are accounted for.

Many of the aforementioned carbon neutrality concepts rely on specific factors, including temporal and spatial boundaries, as well as the origin of the biomass feedstock and alternative scenarios for the utilization or fate of biomass. These elements play a crucial role in determining whether biomass can be considered carbon neutral, as they influence the

assessment of emissions, carbon sequestration, and overall environmental impact. Temporal boundaries, such as the time frame for carbon reabsorption, are critical, with debates on whether to start accounting at harvesting or planting. Longer time frames may offset carbon debt through regrowth but risk delaying climate mitigation. Spatial boundaries, whether plot-level or landscape-level, also impact carbon neutrality assessments, with the latter being preferred for capturing broader carbon dynamics. The choice of biomass feedstock for bioenergy production plays a critical role in climate change mitigation efforts. Forest residues and sawmill waste are widely regarded as favorable feedstocks because they utilize materials that would otherwise decay and release carbon, offering a sustainable way to reduce emissions. Alternative scenarios evaluate the environmental impact of bioenergy production by comparing it to counterfactual situations where biomass is utilized to enhance carbon stocks. However, to comprehensively assess all climate change mitigation options, it is essential to account for emissions linked to the use of alternative energy sources in these scenarios.

During the scoping review, authors confirmed the lack of a common definition of carbon neutrality and that the majority of analyzed articles use more than one carbon neutrality concept. The authors concluded that carbon neutrality is not an inherent property of biomass. The overall idea is that when discussing climate change mitigation, it is more effective to avoid simply labeling a specific biomass source as carbon neutral. Instead, a comprehensive evaluation of the entire bioenergy production process should be conducted, using a transparent and systematic assessment framework to account for overall emissions. This can be accomplished by utilizing LCA approach.

Life cycle assessment

LCA is a method used to analyze the environmental impact associated with a product/system throughout its life cycle. Utilizing “Principles and Framework” and “Requirements and

Guidelines” for LCA, established by International Organization for Standardization (ISO 2006a, ISO 2006b), authors conducted an LCA study in order to assess the emissions related to the production of heat energy from combustion of wooden pellets derived from sawmill residues in the Northwest region of Russia. The facility is located in the suburban industrial area approximately fifteen km away from the city of Arkhangelsk and covers over 100 ha. The facilities’ main activities are lumber and pellet production. The number of pellets produced annually exceeds 100 thousand tons. This region was selected because it is a major hub for forestry and timber processing with several pellet production mills located in the area. The study describes pellet production from sawmill residues, as this biomass type is the region's primary source for biofuel generation.

Goal and scope definition

The main goal of the study was to evaluate the environmental impact of heat generated (1MJ) at a domestic level via combustion of wood pellets produced from sawmill residues in the Arkhangelsk area in the Northwest of Russia. OpenLCA software version 2.1.1 (openlca.org) and Ecoinvent 3.4 database (Ecoinvent 2017) were used for this project.

Life cycle inventory

A gate-to-grave approach of LCA was applied with system boundaries consisting of the following steps: 1) pellet production; 2) pellet distribution; and 3) thermal energy production and waste disposal. Figure 13 presents the boundaries of the bioenergy production system.

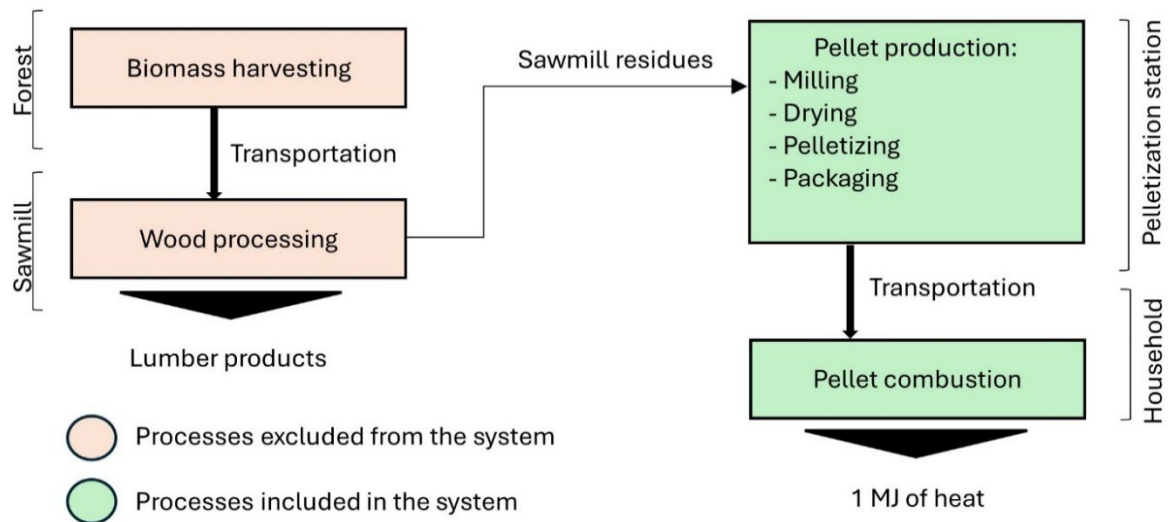


Figure 13. Flow chart of the system boundaries of heat production.

The study simulates a scenario in which biomass (in the form of sawdust, chips and shavings) used for bioenergy production is collected at the sawmill as a by-product of lumber industry activities. The biomass is used to produce wood pellets at the same facility where it was collected. It is important to note that at the pellet production facility in question, biomass is used for heat generation during the drying step of pelletization process. Primary data regarding pellet production was provided by the Executive Director of the “Russian Pellet Union” Association. After pellets are produced, they are stored at the storage unit within the pellet production facility. The pellet producer in question has the capability to sell pellets directly to end-users, bypassing the need for a distributor. The scenario was simulated in which pellets were purchased online directly from the pellet producer and delivered by a light truck (<3.5 t) over the distance of 20 km to the household where they are combusted in a 25kW furnace for heat generation. This distance of 20 km was selected as it represents the approximate radius encompassing the rural communities surrounding the pellet production facility. In this region, most rural residents live in private homes, with district heating plants supplying only 20% of the heat demand. The model therefore assumes the economic and technical feasibility of using locally produced pellets for residential heating.

Life cycle impact assessment

The ReCiPe 2016 (H) Midpoint v. 1.1 (HUIJBREGTS et al. 2017) was used as a method for quantitative life cycle impact assessment across the following impact categories: global warming (GW), freshwater eutrophication (FE), marine eutrophication (ME), terrestrial acidification (TA), fine particulate matter formation (FPMF), fossil resource scarcity (FRS), ozone formation (human health; OFHH), ozone formation (terrestrial ecosystems; OFTE).

The environmental impact of each impact category as well as the results of the sensitivity analyses are presented in Table 6.

Table 6. Sensitivity analysis results: production of 1 MJ of heat by impact category. Baseline scenario – transportation distance of 20km with waste biomass used for drying; sensitivity analysis 2 – natural gas used for drying; sensitivity analysis 3 – transportation distance of 100 km.

Impact categories	Unit	Baseline scenario	Sensitivity analysis 2	Sensitivity analysis 3
		Amount	Amount	Amount
Global warming (GW)	kg CO ₂ eq	0.015	0.016	0.025
Freshwater eutrophication (FE)	kg P eq	5.85e-6	5.82e-6	7.69e-6
Marine eutrophication (ME)	kg N eq	1.42e-6	1.42e-6	1.55e-6
Terrestrial acidification (TA)	kg SO ₂ eq	8.29e-5	8.20e-5	0.00011
Fine particulate matter formation (FPMF)	kg PM _{2.5} eq	7.35e-5	7.22e-5	8.81e-5
Fossil resource scarcity (FRS)	kg oil eq	0.0045	0.0047	0.0075
Ozone formation (human health; OFHH)	kg NO _x eq	0.00012	0.00011	0.00016
Ozone formation (terrestrial ecosystems; OFTE)	kg NO _x eq	0.00012	0.00012	0.00016

GW is identified as the most critical impact category due to its direct link to climate change, with emissions of 0.015 kg CO₂ eq per MJ of heat. These findings align with existing literature, which highlights similar trends in GW emissions for bioenergy systems using waste biomass

and low transportation distances. A review conducted by Martin-Gamboa et al. (2020) showed that GW values from 0 to 0.025 g CO₂ eq use waste biomass as feedstock for pellet production, utilize biomass for drying during pelletization process, assume carbon neutrality of biomass for biogenic emissions, and have low transportation distances. In the LCA in question, pellet production contributes 44.4% of GW emissions, while combustion and waste disposal account for 40.6%, and transportation contributes 15%. Other impact categories include FE (5.85e-6 kg P eq), ME (1.42e-6 kg N eq), TA (8.29e-5 kg SO₂ eq), FPMF (7.35e-5 kg PM_{2.5} eq), FRS (0.0045 kg oil eq), and OF (0.00012 kg NO_x eq for both OFHH and OFTE).

Overall, combustion and waste disposal are the most impactful stages, contributing significantly to five of the eight impact categories. Pellet production is the most impactful in GW, ME, and FRS impact categories, while transportation has a minimal impact, never exceeding 17% in any category.

Sensitivity analyses were conducted to assess the robustness of the results. The first analysis compared the ReCiPe 2016 and ILCD 2011 LCIA methods, revealing similar trends except for marine eutrophication, where methodological differences led to varying results. The second sensitivity analysis evaluated the use of natural gas instead of sawmill residues for drying during pellet production. This change increased GW and fossil resource scarcity emissions by 6.5% and 4.3%, respectively. The third analysis examined the effect of increasing transportation distance from 20 km to 100 km, resulting in significant emission increases across all categories, particularly GW and fossil resource scarcity, which rose by 50%.

The study underscores the importance of optimizing pellet production and transportation to minimize environmental impacts. Combustion and waste disposal remains the most impactful stage, while pellet production significantly influences GW and fossil resource scarcity. Transportation, though less impactful, becomes a critical factor over longer distances.

Conclusions

The results of the study demonstrate that forest-based biomass for bioenergy production is a viable option for climate change mitigation, particularly when compared to fossil fuels. While this study does not directly compare emissions from bioenergy to fossil fuel combustion, existing research supports the efficacy of biofuels. For instance, DIAS et al. (2017) found that pellets used for bioenergy produce significantly fewer emissions (14 g CO₂ eq/MJ heat) compared to natural gas (75 g CO₂ eq/MJ heat) or light fuel oil (87 g CO₂ eq/MJ heat). Similarly, WANG et al. (2017) highlighted that heat generation from pellet combustion results in approximately 94% lower emissions (11.8 g CO₂ eq/MJ heat) compared to coal. These findings underscore the potential of forest biomass as a sustainable alternative to fossil fuels. However, the carbon neutrality of forest biomass is not inherent and depends on various factors, including temporal and spatial boundaries, and bioenergy production practices (e.g. feedstock type, conversion technology, transportation, etc.). The study emphasizes the importance of optimizing production and transportation processes to minimize environmental impacts. Future research should focus on testing specific factors that underpin carbon neutrality concepts, such as temporal and spatial boundaries, as well as exploring alternative biomass feedstock options. Labeling forest biomass as carbon neutral is potentially counterproductive, as this status can be justified using different, often conflicting, carbon accounting concepts. For bioenergy to be truly acceptable, every step of the process must be accounted for via LCA and compared to alternative energy generation methods. Additionally, further investigation into the trade-offs between bioenergy production and forest conservation is essential to ensure sustainable practices. By adopting a transparent and systematic assessment framework, such as LCA, stakeholders can better evaluate the environmental impacts of bioenergy production and make informed decisions that align with climate mitigation goals. Ultimately, forest-based biomass for bioenergy production is a viable option for climate change mitigation, particularly

when compared to fossil fuels. Forest biomass can play a significant role in reducing greenhouse gas emissions, but its sustainability depends on careful management and continuous improvement of production processes.

Chapter 7. Conclusions

The overarching goal of this thesis was to enhance the understanding of the role of forest biomass exploitation for bioenergy production as a CC mitigation option. Three main objectives were identified and addressed in order to achieve this goal.

Objective 1 examined the definition of CN concept of forest biomass in the context of bioenergy production by identifying its underlying assumptions and key limitations. This objective has been fully addressed in Publication I (Chapter 3) through a scoping review. The findings showed that there is no commonly accepted definition of CN. During the process of literature review eight distinct definitions of CN were identified. Analysis of the underlying assumptions of these concepts revealed that CN is not an inherent property of forest biomass, and its CN status is contingent upon specific assumptions and methodological choices, such as the origin of utilized biomass, systemic, temporal, and spatial boundaries, and scenario-based assumptions. Moreover, the results of the scoping review suggest that an LCA methodology should be used to comprehensively assess the environmental impact of biomass exploitation for bioenergy. By addressing Objective 1, this publication supports Hypothesis 1, confirming the lack of a commonly accepted definition of CN, and dependence of the CN status of biomass on specific assumptions and methodological choices.

Objective 2 analyzed the environmental impact of residential heat production in the Northwest region of Russia using wood pellets derived from sawmill residues. This objective has been fully addressed in Publication II through LCA. The findings of this study showed the overall environmental consequences of bioenergy production across several impact categories, additionally analyzing the most impactful stages in the life cycle of the bioenergy system. Moreover, our results confirmed Hypothesis 3 by demonstrating that changes in the supply chain, such as transportation distance and utilization of natural gas instead of residue biomass

for drying during the pellet production stage, can affect the overall environmental performance of the bioenergy system. Additionally, by addressing Objective 2, our LCA study supports Hypothesis 2, which states that forest biomass used for bioenergy production within specific system boundaries has low environmental impact and can be considered as a CC mitigation option. While our study does not include a direct comparison to fossil fuels, emissions related to the bioenergy production in our study consistently fall within the lowest quartile of published comparable bioenergy LCA studies (Martin-Gamboa et al., 2020) suggesting favorable CC mitigation potential. Furthermore, our findings show more positive environmental impacts when compared to the emissions resulting from utilization of fossil fuels observed in other LCA studies (Dias et al., 2017; Wang et al., 2017).

Objective 3 systematically identified the existing methods for monetary evaluation of ES by conducting a scoping review. This objective has been fully addressed in Publication III (Chapter 5) via scoping review. The results of the review supported Hypothesis 4 by distinguishing over twenty monetary valuation approaches. The findings revealed the inconsistencies in the terminology of methods used in literature (e.g., “benefit transfer” vs “value transfer”; “damage cost avoided” vs “averting cost”). Moreover, hybridization of monetary valuation methods was observed (“replacement cost” used as an umbrella term for “substitute cost”). To address terminology inconsistencies and hybridization, future studies should focus on creating a standardized terminology for ES assessment methods and critically evaluating their applicability to bioenergy systems – including comparative analysis of their advantages and disadvantages.

This thesis has identified multiple avenues for future research that warrant further exploration. To further test the viability of forest biomass for bioenergy production, Publication II suggests several directions for potential LCA projects that can build upon the current study, such as (1) evaluation of different transportation logistics (distance and mode), (2) examination of diverse

biofuels used for bioenergy, (3) assessment of alternative biomass feedstocks, and (4) analysis of different conversion technologies. Additionally, the monetary valuation methods identified in Publication III can complement environmental impact assessments by conducting an economic evaluation of the bioenergy production scenario presented in the LCA study. This can be achieved by applying various direct and indirect market valuation approaches (e.g., market pricing, replacement cost, avoided cost) to quantify monetary benefits and trade-offs associated with utilization of forest biomass for bioenergy. Carrying out the proposed projects will enable stakeholders and policymakers to make more informed decisions regarding utilization of forest biomass for bioenergy, considering both environmental impacts and economic viability.

This thesis fully addressed its overarching objective by analyzing the conceptual foundations and limitations of CN of forest biomass (Publication I), performing an environmental assessment of a specific bioenergy production scenario (Publication II), and providing a set of tools for monetary valuation of bioenergy (Publication III) that bridge the gap between environmental analysis and policy implementation. This thesis contributes to the current state of knowledge by systematically examining the concept of CN of forest biomass, demonstrating the need for the LCA approach as a method for comprehensive analysis of biomass exploitation. Additionally, it provides the first comprehensive LCA of small-scale heat generation via combustion of pellets produced from sawmill residues in Russia, addressing a gap in regional bioenergy environmental assessment. Moreover, the scoping review identified and described over 20 monetary evaluation approaches, revealing the terminology inconsistencies and highlighting the necessity for a standardized framework.

This thesis confirmed the overarching hypothesis. Exploitation of forest biomass for bioenergy can contribute to CC mitigation, but its degree sustainability depends on rigorous assessment. This work provides the following key takeaways:

- The CN status of biomass is context specific and depends on the origin of feedstock, as well as temporal and spatial boundaries. Rigorous testing of the CN status of utilized biomass via the LCA methodology is required.
- The CC mitigation potential of bioenergy production is determined by the combined impacts of decisions made at each step of the life cycle. Strategic supply-chain optimization (selection of efficient transportation modes, minimization of transportation distances, sustainable sourcing of biomass materials) will minimize the overall environmental impact.
- The effective implementation of climate sensitive policies necessitates standardized monetary valuation methods to complement the environmental assessment, as current terminology has inconsistencies.

The findings of this thesis provide a methodological framework and empirical evidence to support sustainable bioenergy development for CC mitigation, while highlighting the need for harmonized environmental and economic assessment.

Summary

This thesis is presented as a compilation of four articles alongside an introduction, methodology, and conclusion sections. These articles include: 1) Carbon neutrality (CN) of forest biomass for bioenergy: a scoping review (Chapter 3); 2) Life cycle assessment (LCA) of residential heat production from wood pellet combustion in the Northwest region of Russia (Chapter 4); 3) Methods for monetary valuation of ecosystem services: a scoping review (Chapter 5); and 4) Is forest biomass used for bioenergy production a viable climate change mitigation option? (Chapter 6).

The overarching goal of the thesis is to advance the understanding of the role of forest biomass exploitation for bioenergy production as an option for climate change (CC) mitigation. The key objectives are to analyze the definition and assumptions of carbon neutrality of forest biomass; assess the environmental impact of bioenergy production; and to identify monetary valuation methods for ecosystem services (ES) for future assessment of bioenergy production systems. Each key objective as well as the overarching goal are covered by a dedicated article and chapter.

The general hypothesis of this study is that forest biomass can be used for bioenergy production as a measure to combat CC, provided the CN, environmental impacts, and economic value are rigorously evaluated.

Due to the fact that this thesis explores different aspects of forest biomass exploitation and is presented in the form of a compilation of published articles, different methodological approaches were utilized to conduct these studies. The manuscripts presented in Chapter 3 and Chapter 5 follow a scoping review methodology, whereas Chapter 4 utilizes a standard LCA approach. The publication presented in Chapter 6 adopts the short communication format, which employs a specific concise structure, and uses a qualitative synthesis approach.

The results of the four studies successfully addressed the core objectives of this thesis, demonstrating that:

No universally accepted definition of CN exists, with eight distinct conceptual interpretations identified in current literature.

CN status depends on methodological choices, including biomass origin, system boundaries, and scenario assumptions, confirming that CN is not an inherent property of forest biomass; LCA is recommended for comprehensive environmental evaluation.

Residential heat production from wood pellets in Northwest Russia demonstrated low environmental impact and a significant influence of supply chain decisions (e.g., transport distance, drying methods) on environmental performance.

A scoping review of ecosystem service valuation revealed over twenty monetary valuation methods, highlighting inconsistencies in terminology (e.g., "benefit transfer" versus "value transfer") and widespread hybridization of approaches.

Standardization of ES valuation methods is needed to improve economic assessments of bioenergy systems and to inform policy decisions.

The general conclusion of this thesis confirms the general hypothesis. Exploitation of forest biomass for bioenergy can contribute to CC mitigation, but its degree of sustainability depends on rigorous assessment. This thesis provides a methodological foundation for sustainable bioenergy development, bridging environmental analysis and policy implementation while emphasizing the need for integrated assessment approaches.

Souhrn

Tato práce je prezentována jako kompilace čtyř článků doplněných úvodem, metodikou a závěrem. Těmito články jsou: 1) Carbon neutrality of forest biomass for bioenergy: a scoping review (kapitola 3); 2) Life cycle assessment of residential heat production from wood pellet combustion in the Northwest region of Russia (kapitola 4); 3) Methods for monetary valuation of ecosystem services: a scoping review (kapitola 5); a 4) Je lesní biomasa používána k výrobě bioenergie uhlíkově neutrální? (kapitola 6).

Hlavním cílem této práce bylo prohloubit pochopení role využívání lesní biomasy pro výrobu bioenergie jako možnosti mitigace změny klimatu (ZK). Hlavní úkoly práce jsou analyzovat definici a předpoklady uhlíkové neutrality lesní biomasy; posoudit dopad výroby bioenergie na životní prostředí; a identifikovat metody peněžního oceňování ekosystémových služeb (ES) pro budoucí hodnocení systémů výroby bioenergie. Každý z hlavních úkolů je zpracován formou článku obsaženém v příslušné kapitole.

Základní hypotézou této studie je, že lesní biomasa může být využita k výrobě bioenergie jako opatření k boji proti ZK, za předpokladu pečlivého posouzení její uhlíkové neutrality (UN), dopadů na životní prostředí a ekonomické hodnoty.

Vzhledem k tomu, že tato práce zkoumá různé aspekty využívání lesní biomasy a je prezentována ve formě kompilace publikovaných článků, byly k provedení těchto studií použity různé metodické přístupy. Publikace prezentované v kapitole 3 a kapitole 5 využívají metodu „scoping review“, zatímco kapitola 4 využívá standardní přístup metody posuzování životního cyklu (LCA – Life Cycle Assessment). Publikace uvedená v kapitole 6 je zpracována formou krátkého sdělení a vyznačuje se specifickou, výstižnou strukturou a použitím přístupu kvalitativní syntézy.

Výsledky těchto studií úspěšně splnily hlavní cíle této práce a prokázaly, že:

- Neexistuje žádná univerzálně přijímaná definice UN, přičemž bylo identifikováno osm odlišných koncepčních přístupů.
- Status UN závisí na zvoleném metodickém přístupu, zejména na původu biomasy, vymezení systému a předpokladech scénářů, což potvrzuje, že UN není inherentní vlastností lesní biomasy a pro komplexní environmentální hodnocení se proto doporučuje využívat LCA.
- Výroba tepla pro domácnosti z dřevěných pelet v severozápadním Rusku ukázala nízký dopad na životní prostředí a dále, že rozhodnutí v dodavatelském řetězci (např. přepravní vzdálenost a metody sušení) významně ovlivnila environmentální dopady.
- Scoping review oceňování ES odhalil více než dvacet metod peněžního oceňování, a zároveň upozornil na nesrovnalosti v terminologii (např. „přenos přínosů“ versus „přenos hodnoty“) a na široce rozšířené kombinování různých přístupů.
- Standardizace metod oceňování ES je nezbytná pro zlepšení ekonomického hodnocení bioenergetických systémů a podpoření informovaných politických rozhodnutí.

Obecný závěr této práce potvrzuje základní hypotézu. Využívání lesní biomasy pro bioenergii může přispět k mitigaci ZK, ale stupeň její udržitelnosti závisí na důkladném posouzení. Tato práce poskytuje metodický základ pro udržitelný rozvoj bioenergie, propojuje environmentální analýzu a implementaci politik a zároveň zdůrazňuje potřebu integrovaných přístupů k hodnocení.

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